*MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, UP/DOWN, BINARY, KEY CODE BBA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic binary up/down counter module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.364 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BBA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/30-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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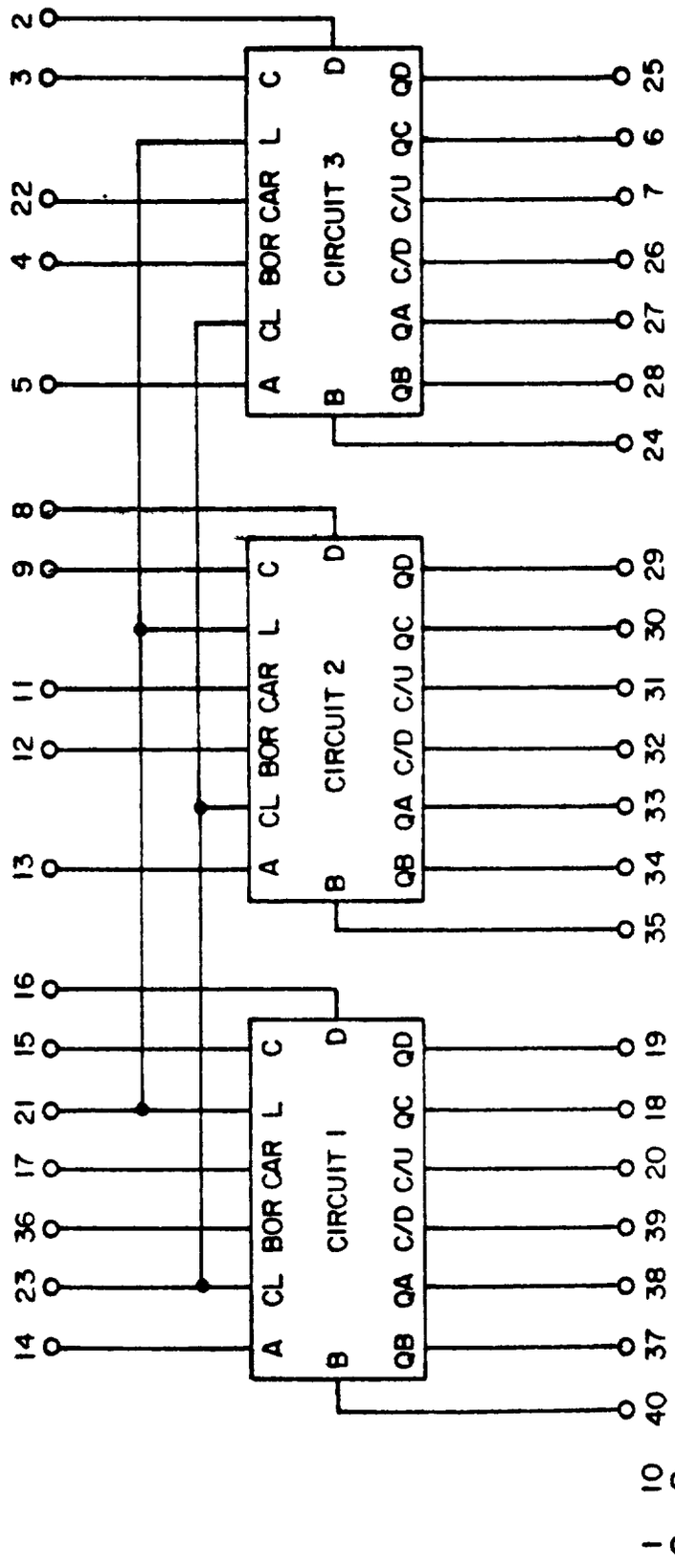


FIGURE 1. BBA module function diagram, binary up/down counter.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A18	V_{OUT} (low loaded)	III	S	10	800	10	800	10	800	10	800	mV
B1-B18	Input breakdown current (A, B, C, D, C/U, C/D)	IV	S	0	1	0	1	0	1	0	1	mA
B19, B20	Input breakdown current (CL, L)	IV	S	0	3	0	3	0	3	0	3	mA
C	Isolation	V	S	10	-	1	-	10	-	1	-	MΩ
D1-D20	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E18	Input current, low level (A, B, C, D, C/U, C/D)	VII	X	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	mA
E19, E20	Input current, low level (CL, L)	VII	X	-0.30	-5.19	-0.30	-5.19	-0.30	-5.19	-0.30	-5.19	mA
F1-F18	Input current, high level (A, B, C, D, C/U, C/D)	VII	X	0	40	0	40	0	40	0	40	µA
F19, F20	Input current, high level (CL, L)	VII	X	0	120	0	120	0	120	0	120	µA
G1-G18	Output short circuit current	VII	X	-18	-65	-18	-65	-18	-65	-18	-65	mA
H1-H18	Output voltage, low level	VIII	X	0	400	0	400	0	400	0	400	mV
J1-J18	Output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K53	Functional	IX	X	Test vectors must produce output vectors as specified in table IX.								
L1, L2	Power supply current, high level and low level	X	X	50	306	50	306	50	306	50	306	mA
M1-M6	Propagation delay time (t_{PHL})	XI	S	3	35	3	35	3	40	3	40	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1-N6	Propagation delay time (t_{PLH})	XI	S	3	35	3	35	3	40	3	40	ns
P1-P12	Propagation delay time (t_{PLH})	XII	S	3	47	3	47	3	51	3	51	ns
Q1-Q12	Propagation delay time (t_{PHL})	XII	S	3	56	3	56	3	60	3	60	ns
R1-R12	Propagation delay time (t_{PLH})	XII	S	3	47	3	47	3	51	3	51	ns
S1-S12	Propagation delay time (t_{PHL})	XII	S	3	56	3	56	3	60	3	60	ns
T1-T18	Transition time (t_{TLH})	XIII	S	10	45	10	45	10	50	10	50	ns
U1-U18	Transition time (t_{THL})	XIII	S	5	20	5	20	5	25	5	25	ns
V1-V12	Propagation delay time (t_{PLH})	XIV	S	3	49	3	49	3	57	3	57	ns
W1-W12	Propagation delay time (t_{PHL})	XIV	S	3	49	3	49	3	57	3	57	ns
X1-X12	Propagation delay time (t_{PHL})	XIV	S	3	44	3	44	3	50	3	50	ns
Y1-Y6	Maximum counting frequency check (minimum count pulse width check)	XV	S	1,070	1,140	1,070	1,140	1,070	1,140	1,070	1,140	ns
Z1-Z12	Setup time (data to load) and minimum load pulse width check	XVI	S	186	214	186	214	186	214	186	214	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
AA	Power supply filter capacitance	XVII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
BB1-BB12	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	mV
CC1-CC12	Output crosstalk, low level	XVIII	Q	0	1	0	1	0	1	0	1	V
CC13-CC24	Output crosstalk, duration	XVIII	Q	0	400	0	400	0	400	0	400	mV
DD	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	mV
EE1-EE18	Input capacitance (A, B, C, D, C/U, C/D)	XX	Q	2	15	2	15	2	15	2	15	pF
EE19, EE20	Input capacitance (CL, L)	XX	Q	4	30	4	30	4	30	4	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through EE shall be performed in any order with the following exceptions. If test conditions A, B, C, or G are part of a sequence of tests, the order of testing shall be as follows: C, A, B, G where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through EE in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through DD shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition AA, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through DD shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through EE shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military specification MIL-M-28787/30 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833643. Part number M28787/30-1 supersedes part numbers 2833643 and M28787/3D.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	1.69 W (outputs not loaded)
Supply voltage	5.0 V dc ±10 percent
Clock inputs	
Counting frequency	DC to 15 MHz maximum
Pulse width	40 ns minimum
Triggering edge	Positive

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Input voltage	
High level	2.0 V minimum, V_{CC} maximum
Low level	-0.5 V minimum, 0.8 V maximum
Input current (data, count up, count down), high level	
(V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	1.0 mA maximum
Input current (clear, load), high level	
(V_{IH} = 2.4 V)	120 µA maximum
(V_{IH} = 5.5 V)	3.0 mA maximum
Input current (data, count up, count down), low level	
(V_{IL} = 0.0 V)	-1.73 mA maximum
(V_{IL} = 0.4 V)	-1.60 mA maximum
Input current (clear, load), low level	
(V_{IL} = 0.0 V)	-5.19 mA maximum
(V_{IL} = 0.4 V)	-4.80 mA maximum
Output voltage	
High level (I_{OH} = -400 µA)	2.4 V minimum
Low level (I_{OL} = 16.0 mA)	0.4 V maximum
Output drive maximum capability	
High level (V_{OH} = 2.4 V)	-400 µA
Low level (V_{OL} = 0.4 V)	16.0 mA
Propagation delay time	
Count up to carry (t_{PLH})	3 ns minimum, 40 ns maximum
Count up to carry (t_{PHL})	3 ns minimum, 40 ns maximum
Count down to borrow (t_{PLH})	3 ns minimum, 40 ns maximum
Count down to borrow (t_{PHL})	3 ns minimum, 40 ns maximum
Count up to Q (t_{PHL})	3 ns minimum, 60 ns maximum
Count down to Q (t_{PHL})	3 ns minimum, 60 ns maximum
Count up to Q (t_{PLH})	3 ns minimum, 51 ns maximum
Count down to Q (t_{PLH})	3 ns minimum, 51 ns maximum
Clear to Q (t_{PHL})	3 ns minimum, 50 ns maximum
Load to Q (t_{PLH})	3 ns minimum, 57 ns maximum
Load to Q (t_{PHL})	3 ns minimum, 57 ns maximum
Data and clear input pulse width	25 ns minimum
Load input pulse width	35 ns minimum
Data setup time (before positive transition of load pulse)	35 ns minimum
Data hold time (after positive transition of load pulse)	5 ns minimum

* 6.5 Module function. This module contains three 4-bit synchronous binary up/down counters. Each counter has parallel inputs and outputs and carry and borrow outputs. The counters have common clear and load inputs and independent count up and count down clock inputs. The functional diagram is illustrated on figure 1. All three counters are fully programmable; that is, the outputs may be preset to any state by entering the desired data at the DATA inputs while the LOAD input is low. The output will change to agree with the DATA inputs independently of the count pulses. The CLEAR input forces all outputs to the low level when a high level is applied. The CLEAR function is independent of the COUNT and LOAD inputs. Both BORROW and CARRY outputs are available to cascade both the up and down counting functions. The BORROW output produces a pulse equal in width to the COUNT DOWN input when the counter underflows, and the CARRY output produces a pulse equal in width to the COUNT UP input when the counter overflows.

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6.5.1 Input to output relationship. Each counter has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 4/

	Inputs								Outputs			
	Clear	Load	Data				Count up	Count down	Q_A	Q_B	Q_C	Q_D
			A	B	C	D						
Asynchronous operation	H L	X L	X A	X B	X C	X D	X X	X X	L A	L B	L C	L D
Synchronous operation	L L	H H	X X	X X	X X	X X	H 5/	H 5/	Count down Count up			

1/ X = Unspecified (either high level or low level).

2/ CARRY = $\overline{Q_A \cdot Q_B \cdot Q_C \cdot Q_D \cdot C/D \cdot \overline{C/U}}$ (CARRY is ACTIVE low).

3/ BORROW = $\overline{\overline{Q_A} \cdot \overline{Q_B} \cdot \overline{Q_C} \cdot \overline{Q_D} \cdot C/U \cdot \overline{C/D}}$ (BORROW is ACTIVE low).

4/ L = low level and H = high level.

5/ Function is synchronous with the positive-going edge of the count pulse.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

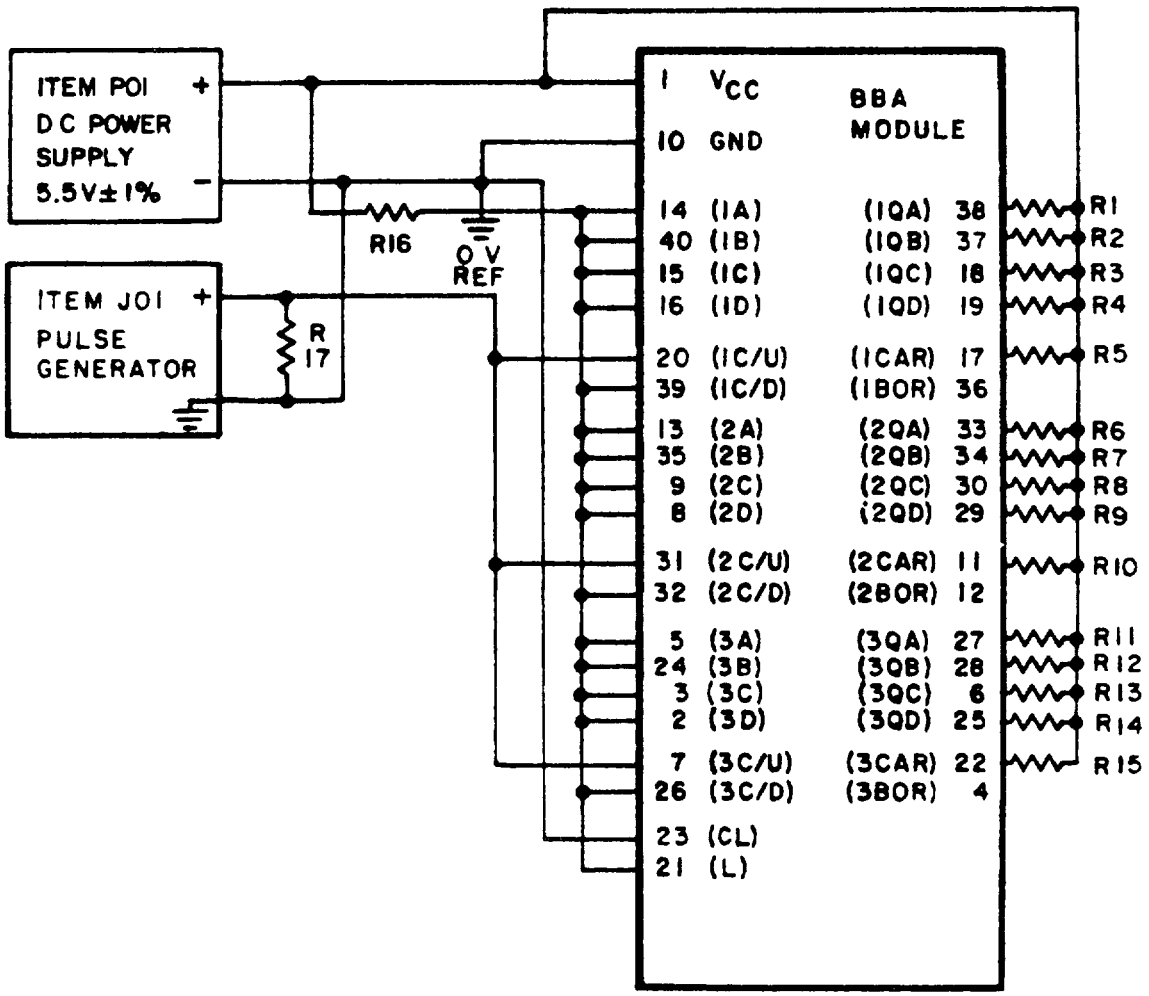
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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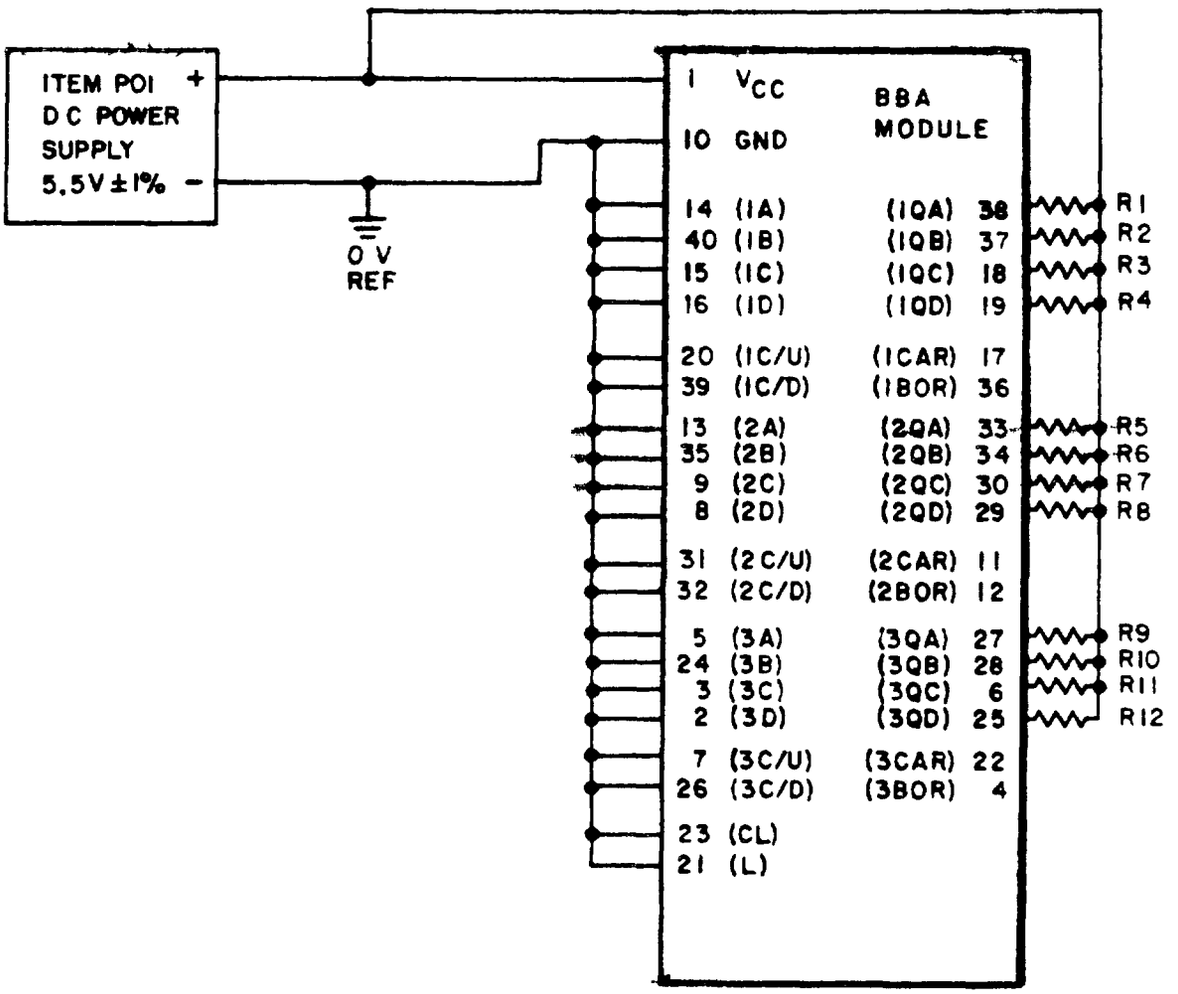
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NOTES:
1. R1-R15 = 316Ω ±1%, 1/4 W.
2. R16 = 1.0 kΩ ±10%, 1/8 W.
3. R17 = 51Ω ±5%, 1/2 W.
4. Item JO1 setup:
   PW = 500 ±50 ns.
   PRP = 1.0 ±0.1 µs.
   $t_{TLH} = t_{THL}$ = 50 ±20 ns.
   PA = 4.0 ±0.2 V.

* FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R12 = 316Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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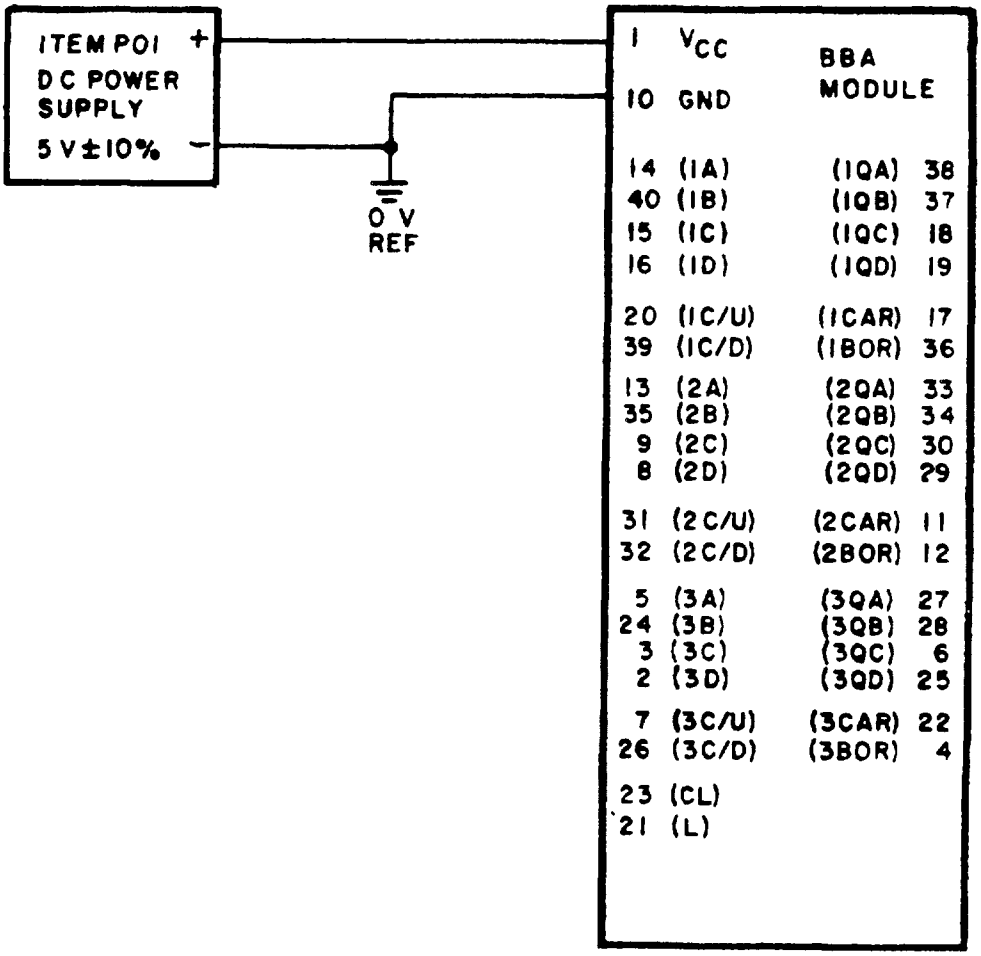


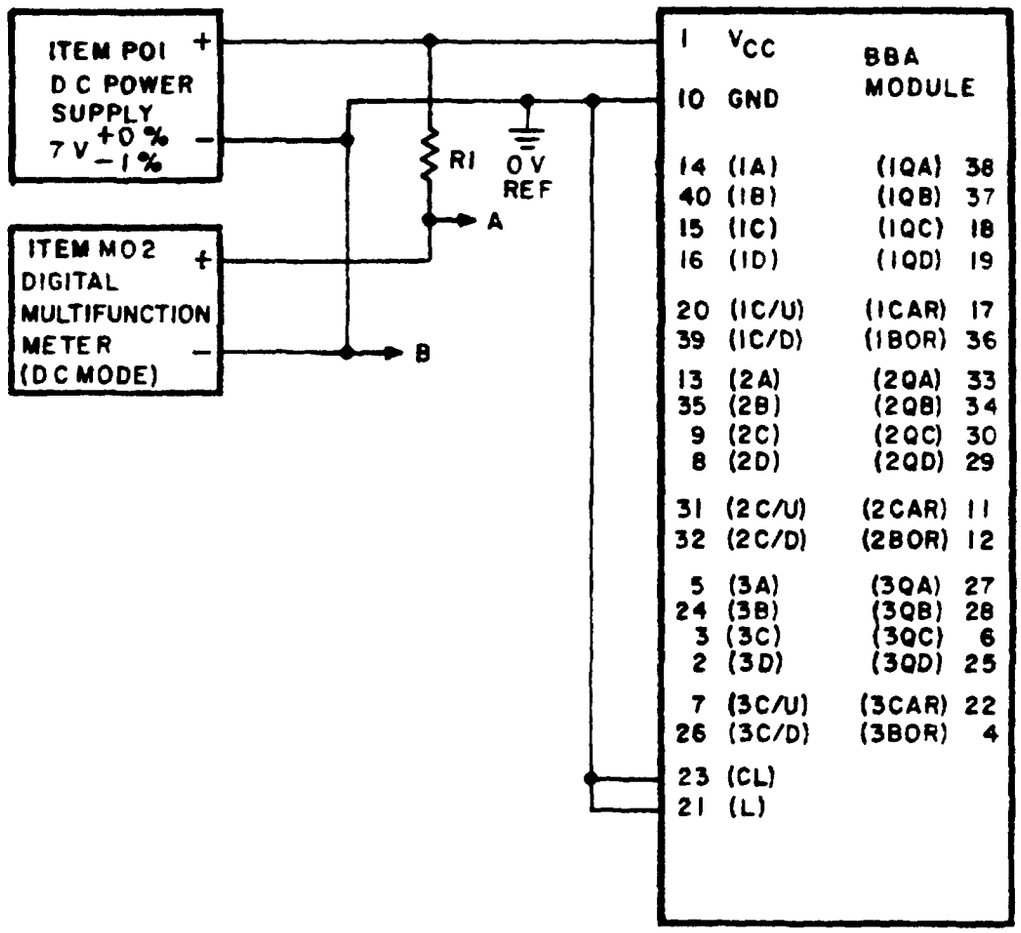
FIGURE 4. Vibration test circuit.

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*TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	1QA V_{OUT} (low loaded)	38	14	10	800	mV	5
A2	1QB "	37	40	"	"	"	"
A3	1QC "	18	15	"	"	"	"
A4	1QD "	19	16	"	"	"	"
A5	1CAR "	17	20	"	"	"	"
A6	1BOR "	36	14,15,16, 39,40	"	"	"	"
A7	2QA "	33	13	"	"	"	"
A8	2QB "	34	35	"	"	"	"
A9	2QC "	30	9	"	"	"	"
A10	2QD "	29	8	"	"	"	"
A11	2CAR "	11	31	"	"	"	"
A12	2BOR "	12	8,9,13, 32,35	"	"	"	"
A13	3QA "	27	5	"	"	"	"
A14	3QB "	28	24	"	"	"	"
A15	3QC "	6	3	"	"	"	"
A16	3QD "	25	2	"	"	"	"
A17	3CAR "	22	7	"	"	"	"
A18	3BOR "	4	2,3,5,24,26	"	"	"	"

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NOTE: R1 = 316Ω ±1%, 1/4 W.

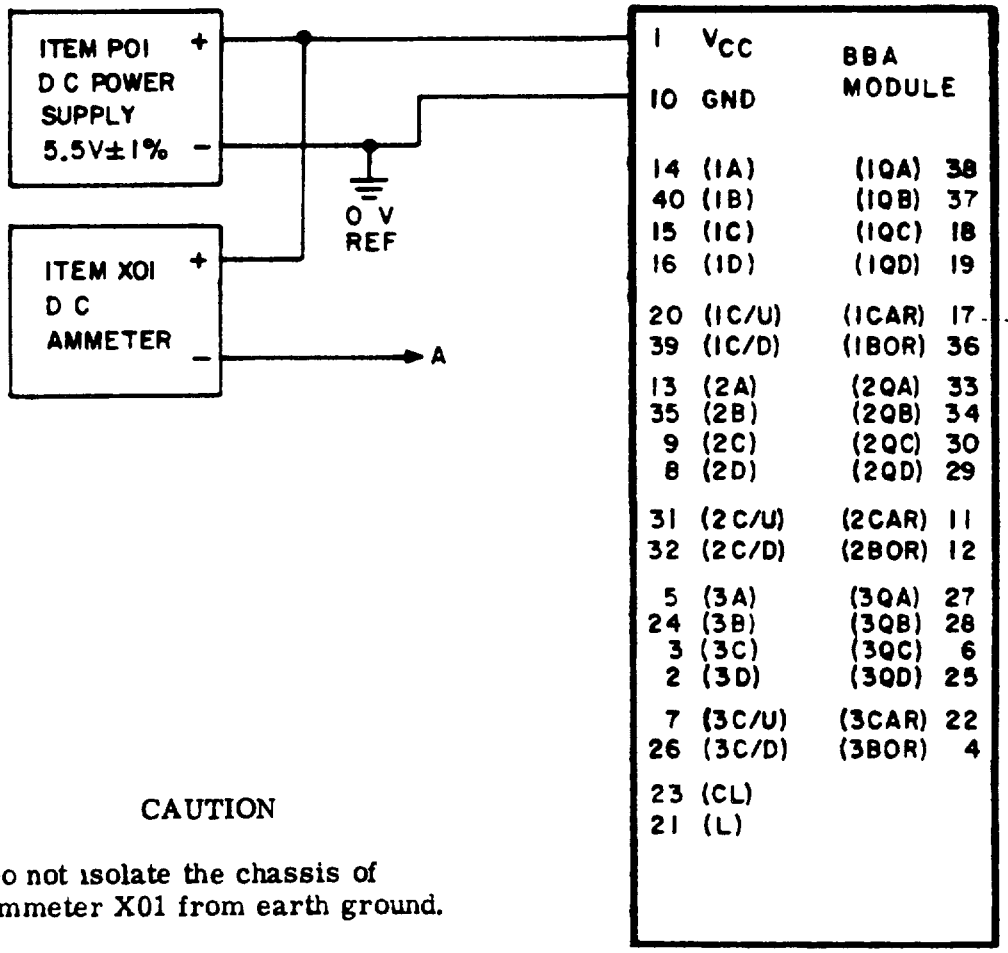
* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	1A Input breakdown current	14	0	1	mA	6
B2	1B "	40	"	"	"	"
B3	1C "	15	"	"	"	"
B4	1D "	16	"	"	"	"
B5	1C/U "	20	"	"	"	"
B6	1C/D "	39	"	"	"	"
B7	2A "	13	"	"	"	"
B8	2B "	35	"	"	"	"
B9	2C "	9	"	"	"	"
B10	2D "	8	"	"	"	"
B11	2C/U "	31	"	"	"	"
B12	2C/D "	32	"	"	"	"
B13	3A "	5	"	"	"	"
B14	3B "	24	"	"	"	"
B15	3C "	3	"	"	"	"
B16	3D "	2	"	"	"	"
B17	3C/U "	7	"	"	"	"
B18	3C/D "	26	"	"	"	"
B19	CL "	23	"	3	"	"
B20	L "	21	"	3	"	"

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

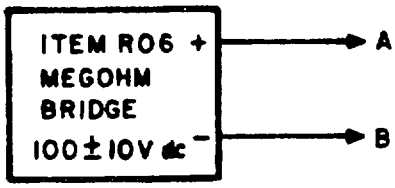
FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Isolation	All pins	Frame	10	∞	MΩ	7

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1	V_{CC}	BBA MODULE	
10	GND		
14	(1A)	(1QA)	38
40	(1B)	(1QB)	37
15	(1C)	(1QC)	18
16	(1D)	(1QD)	19
20	(1C/U)	(1CAR)	17
39	(1C/D)	(1BOR)	36
13	(2A)	(2QA)	33
35	(2B)	(2QB)	34
9	(2C)	(2QC)	30
8	(2D)	(2QD)	29
31	(2C/U)	(2CAR)	11
32	(2C/D)	(2BOR)	12
5	(3A)	(3QA)	27
24	(3B)	(3QB)	28
3	(3C)	(3QC)	6
2	(3D)	(3QD)	25
7	(3C/U)	(3CAR)	22
26	(3C/D)	(3BOR)	4
23	(CL)		
21	(L)		

WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe.

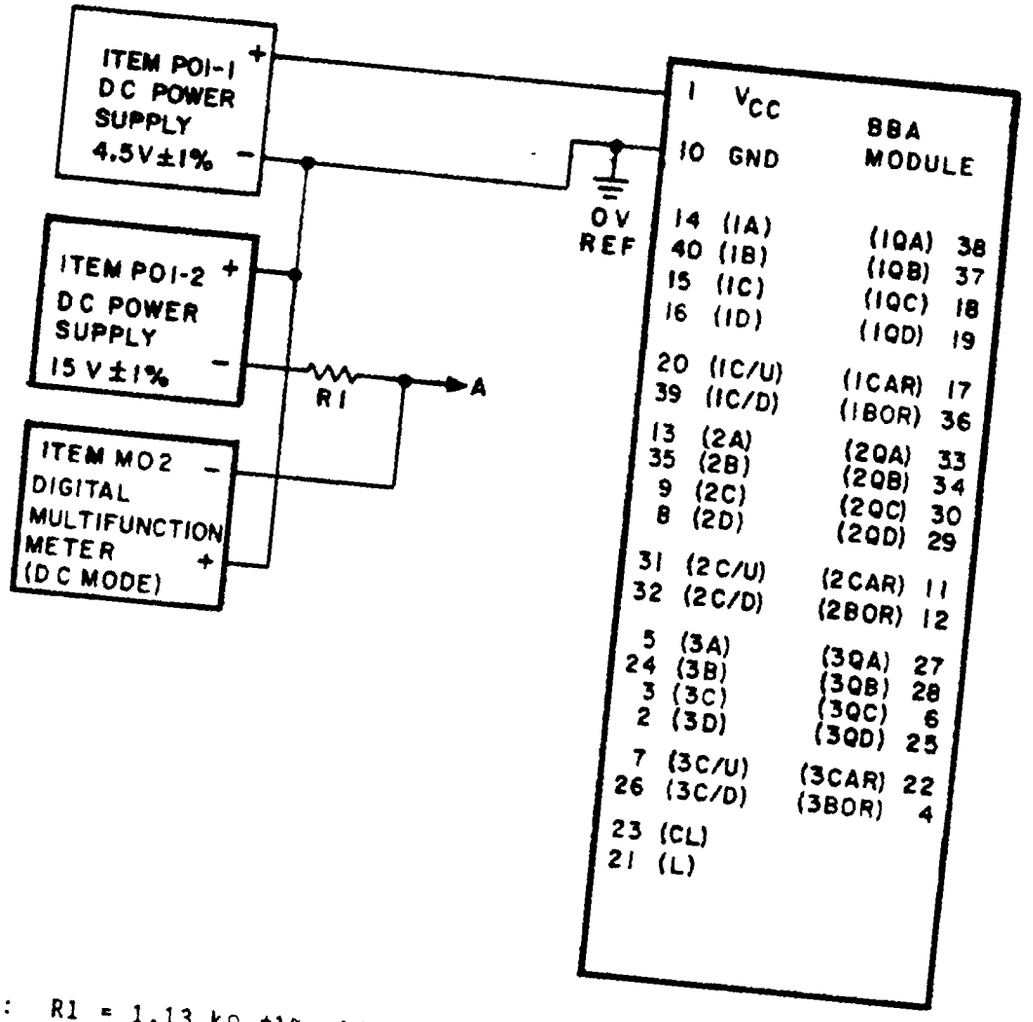
FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D1	1A Input clamp diode voltage	14	-0.4	-1.5	V	8
D2	1B "	40	"	"	"	"
D3	1C "	15	"	"	"	"
D4	1D "	16	"	"	"	"
D5	1C/U "	20	"	"	"	"
D6	1C/D "	39	"	"	"	"
D7	2A "	13	"	"	"	"
D8	2B "	35	"	"	"	"
D9	2C "	9	"	"	"	"
D10	2D "	8	"	"	"	"
D11	2C/U "	31	"	"	"	"
D12	2C/D "	32	"	"	"	"
D13	3A "	5	"	"	"	"
D14	3B "	24	"	"	"	"
D15	3C "	3	"	"	"	"
D16	3D "	2	"	"	"	"
D17	3C/U "	7	"	"	"	"
D18	3C/D "	26	"	"	"	"
D19	CL "	23	"	"	"	"
D20	L "	21	"	"	"	"

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NOTE: R1 = 1.13 kΩ ±1%, 1/8 W.

* FIGURE 8. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Input current (low-level and high-level) and output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	1A Input current, low level	21, 23	Open	14	Open	-0.10	-1.73	mA	9
E2	1B "	"	"	40	"	"	"	"	"
E3	1C "	"	"	15	"	"	"	"	"
E4	1D "	"	"	16	"	"	"	"	"
E5	1C/U "	Open	"	20	"	"	"	"	"
E6	1C/D "	Open	"	39	"	"	"	"	"
E7	2A "	21, 23	"	13	"	"	"	"	"
E8	2B "	"	"	35	"	"	"	"	"
E9	2C "	"	"	9	"	"	"	"	"
E10	2D "	"	"	8	"	"	"	"	"
E11	2C/U "	Open	"	31	"	"	"	"	"
E12	2C/D "	Open	"	32	"	"	"	"	"
E13	3A "	21, 23	"	5	"	"	"	"	"
E14	3B "	"	"	24	"	"	"	"	"
E15	3C "	"	"	3	"	"	"	"	"
E16	3D "	"	"	2	"	"	"	"	"
E17	3C/U "	Open	"	7	"	"	"	"	"
E18	3C/D "	"	"	26	"	"	"	"	"
E19	CL "	"	"	23	"	-0.30	-5.19	"	"
E20	L "	"	"	21	"	-0.30	-5.19	"	"

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1A Input current, high level	15,16,20, 39,40	14	Open	21, 23	0	40	µA	9
F2	1B "	14,15,16, 20,39	40	"	"	"	"	"	"
F3	1C "	14,16,20, 39,40	15	"	"	"	"	"	"
F4	1D "	14,15,20, 39,40	16	"	"	"	"	"	"
F5	1C/U "	14,15,16, 21,23,39, 40	20	"	Open	"	"	"	"
F6	1C/D "	14,15,16, 20,21,23, 40	39	"	Open	"	"	"	"
F7	2A "	8,9,31,32, 35	13	"	21, 23	"	"	"	"
F8	2B "	8,9,13,31, 32	35	"	"	"	"	"	"
F9	2C "	8,13,31, 32,35	9	"	"	"	"	"	"
F10	2D "	9,13,31, 32,35	8	"	"	"	"	"	"
F11	2C/U "	8,9,13,21, 23,32,35	31	"	Open	"	"	"	"
F12	2C/D "	8,9,13,21, 23,31,35	32	"	Open	"	"	"	"

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

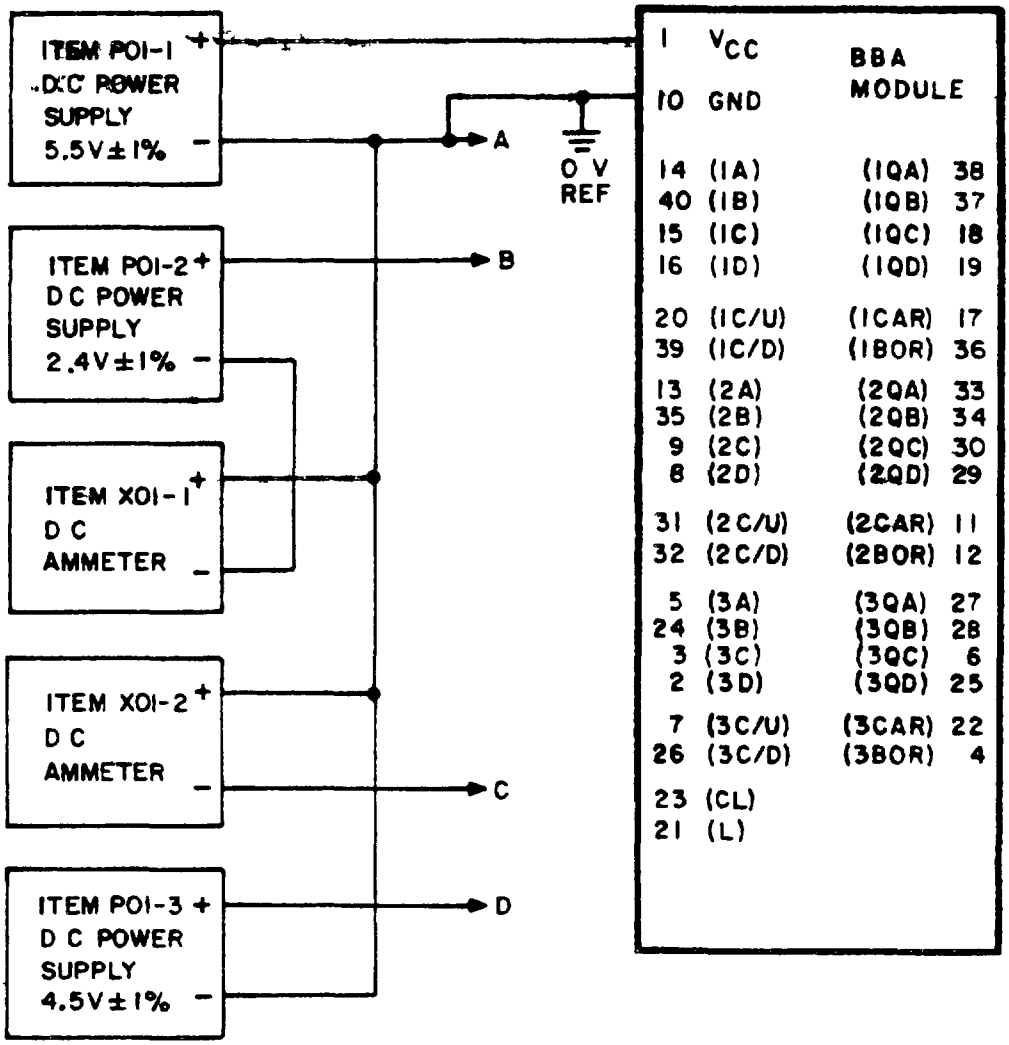
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F13	3A Input current, high level	2,3,7,24, 26	5	Open	21, 23	0	40	µA	9
F14	3B "	2,3,5,7,26	24	"	"	"	"	"	"
F15	3C "	2,5,7,24, 26	3	"	"	"	"	"	"
F16	3D "	3,5,7,24, 26	2	"	"	"	"	"	"
F17	3C/U "	2,3,5,21, 23,24,26	7	"	Open	"	"	"	"
F18	3C/D "	2,3,5,7, 21,23,24	26	"	"	"	"	"	"
F19	CL "	2,3,5,7,8, 9,13,14,15, 16,20,21, 24,26,31, 32,35,39, 40	23	"	"	"	120	"	"
F20	L "	2,3,5,7,8, 9,13,14,15, 16,20,23, 24,26,31, 32,35,39, 40	21	"	"	"	120	"	"
G1	1QA Output short circuit current	21, 23	Open	38	14,20,39	-18	-65	mA	"
G2	1Q8 "	"	"	37	20,39,40	"	"	"	"
G3	1QC "	"	"	18	15,20,39	"	"	"	"
G4	1QD "	"	"	19	16,20,39	"	"	"	"
G5	1CAR "	"	"	17	20,39	"	"	"	"

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TABLE VII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G6	1BOR Output short circuit current	21, 23	Open	36	20,39	-18	-65	mA	9
G7	2QA "	"	"	33	13,31,32	"	"	"	"
G8	2QB "	"	"	34	31,32,35	"	"	"	"
G9	2QC "	"	"	30	9,31,32	"	"	"	"
G10	2QD "	"	"	29	8,31,32	"	"	"	"
G11	2CAR "	"	"	11	31,32	"	"	"	"
G12	2BOR "	"	"	12	31,32	"	"	"	"
G13	3QA "	"	"	27	5,7,26	"	"	"	"
G14	3QB "	"	"	28	7,24,26	"	"	"	"
G15	3QC "	"	"	6	3,7,26	"	"	"	"
G16	3QD "	"	"	25	2,7,26	"	"	"	"
G17	3CAR "	"	"	22	7,26	"	"	"	"
G18	3BOR "	"	"	4	7,26	"	"	"	"

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NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. Only one output shall be shorted at a time, and the measurement taken within 10 seconds.

FIGURE 9. Input current (low-level and high-level) and output short circuit current test circuit for table VII.

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TABLE VIII Output voltage (low-level and high-level) test.

Test condition	Parameter	Connect				Output logic states																		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1QA	1QB	1QC	1QD	1CAR	1BQR	2QA	2QB	2QC	2QD	2CAR	2BQR	3QA	3QB	3QC	3QD	3CAR	3BQR	Min	Max		
H1	1QA Output voltage, low level	2,3,5,8,9,13,14,15,16,21,23,24,26,32,35,39,40	7,20,31	38	Open	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	0	400	mV	10
H2	1QB "	"	"	37	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H3	1QC "	"	"	18	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H4	1QD "	"	"	19	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H5	2QA "	"	"	33	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H6	2QB "	"	"	34	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H7	2QC "	"	"	30	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H8	2QD "	"	"	29	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H9	3QA "	"	"	27	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H10	3QB "	"	"	28	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H11	3QC "	"	"	6	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H12	3QD "	"	"	25	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H13	1BQR "	"	"	36	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H14	2BQR "	"	"	12	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H15	3BQR "	"	"	4	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L	"	"	"	"
H16	1CAR "	7,20,21,23,31	2,3,5,8,9,13,14,15,16,24,26,32,35,39,40	17	"	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	"	"	"	"
H17	2CAR "	7,20,21,23,31	2,3,5,8,9,13,14,15,16,24,26,32,35,39,40	11	"	L	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	"	"	"	"

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TABLE VIII. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Output logic states																		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1 QA	1 QB	1 QC	1 QD	1 CAR	1 BOR	2 QA	2 QB	2 QC	2 QD	2 CAR	2 BOR	3 QA	3 QB	3 QC	3 QD	3 CAR	3 BOR	Min	Max		
H18	3CAR Output voltage, low level	7,20,21, 23,31	2,3,5,8, 9,13,14, 15,16,24, 26,32,35, 39,40	22	Open	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	0	400	mV	10
J1	1QA Output voltage, high level	21,23	2,3,5,7, 8,9,13, 14,15,16, 20,24,26, 31,32,35, 39,40	38	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	2.4	4.5	V	"
J2	1QB "	"	"	37	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J3	1QC "	"	"	18	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J4	1QD "	"	"	19	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J5	2QA "	"	2,3,5,7, 8,9,13, 14,15,16, 20,24,26, 31,32,35, 39,40	33	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J6	2QB "	"	"	34	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J7	2QC "	"	"	30	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J8	2QD "	"	"	29	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J9	3QA "	"	"	27	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J10	3QB "	"	"	28	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J11	3QC "	"	"	6	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J12	3QD "	"	"	25	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J13	1BOR "	"	"	36	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J14	2BOR "	"	"	12	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J15	3BOR "	"	"	4	"	H	H	H	H	H	H	H	h	H	H	H	H	H	H	H	H	H	H	"	"	"	"

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TABLE VIII. Output voltage (low-level and high-level) test - Continued

Test condition	Parameter	Connect				Output logic states																		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1QA	1QB	1QC	1QD	1CAR	1BDR	2QA	2QB	2QC	2QD	2CAR	2BDR	3QA	3QB	3QC	3QD	3CAR	3BDR	Min	Max		
J16	1CAR Output voltage, high level	21,23	2,3,5,7, 8,9,13, 14,15,16, 20,24,26, 31,32,35, 39,40	17	Open	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	2.4	4 5	V	10
J17	2CAR "	"	"	11	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"
J18	3CAR "	"	"	22	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"

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TABLE IX. Functional test

Test condition	Parameter	Connect				Ouput logic states																Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1 Q A	1 Q B	1 Q C	1 Q D	1 C A R	1 B Q R	2 Q A	2 Q B	2 Q C	2 Q D	2 C A R	2 B Q R	3 Q A	3 Q B	3 Q C	3 Q D	3 C A R	3 B Q R	Min	Max		
K1	Functional, parallel load	21,23	2,3,5,7, 8,9,13, 14,15,16, 20,24,26, 31,32,35, 39,40	1/ 2/	Open	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H		3/		10
K2	Functional, clear	21	2,3,5,7, 8,9,13, 14,15,16, 20,23,24, 26,31,32, 35,39,40	"	Open	L	L	L	L	H	H	L	L	L	L	H	H	L	L	L	L	H	H		"		"
K3	Functional, parallel load	21,23	2,3,5,8, 9,13,14, 15,16,24, 26,32,35, 39,40	"	7,20, 31	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H		"		"
K4	"	3,5,9,13, 14,15,16, 21,23,24, 35,40	2,7,8,16, 20,31	"	26,32, 39	L	L	L	H	H	H	L	L	L	H	H	H	L	L	L	H	H	H		"		"
K5	"	2,8,16,21, 23	3,5,9,13, 14,15,24, 26,32,35, 39,40	"	7,20, 31	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H		"		"
K6	"	3,9,15,21, 23,24,35, 40	2,5,7,8, 13,14,16, 20,31	"	26,32, 39	H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H		"		"
K7	"	2,5,8,13, 14,16,21, 23	3,9,15, 24,26,32, 35,39,40	"	7,20, 31	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H		"		"
K8	"	21,23,24, 35,40	2,3,5,7, 8,9,13, 14,15,16, 20,31	"	26,32, 39	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H		"		"
K9	"	2,3,5,8,9, 13,14,15, 16,21,23	24,26,32, 35,39,40	"	7,20, 31	L	H	L	L	H	H	L	H	L	L	H	H	L	H	L	L	H	H		"		"

See footnotes at end of table.

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TABLE IX. Functional test - Continued.

Test condition	Parameter	Connect				Ouput logic states																		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1 Q A	1 Q B	1 Q C	1 Q D	1 CA K	1 BQ K	2 Q A	2 Q B	2 Q C	2 Q D	2 CA K	2 dQ K	3 Q A	3 Q B	3 Q C	3 Q D	3 CA K	3 BQ K	Min	Max		
K10	Functional, parallel load	5,13,14,21, 23,24,35, 40	2,3,7,8, 9,15,16, 20,31	1/ 2/	26,32, 39	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H	H		3/		10
K11	"	2,3,8,9,15, 16,21,23, 24,35,40	5,13,14, 26,32,39	"	7,20, 31	H	L	L	L	H	H	H	L	L	L	H	H	H	L	L	L	H	H		"		"
K12	"	5,13,14,21, 23	2,3,7,8, 9,15,16, 20,24,31, 35,40	"	26,32, 39	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H		"		"
K13	"	2,3,8,9, 15,16,21, 23	5,13,14, 24,26,32, 35,39,40	"	7,20, 31	H	H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H		"		"
K14	"	2,5,8,13, 14,16,21, 23,24,35, 40	3,7,9,15, 20,31	"	26,32, 39	L	L	H	L	H	H	L	L	H	L	H	H	L	L	H	L	H	H		"		"
K15	"	3,9,15,21, 23	2,5,8,13, 14,16,24, 26,32,35, 39,40	"	7,20, 31	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H		"		"
K16	"	2,8,16,21, 23,24,35, 40	3,5,7,9, 13,14,15, 20,31	"	26,32, 39	H	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H		"		"
K17	"	3,5,9,13, 14,15,21, 23	2,8,16, 24,26,32, 35,39,40	"	7,20, 31	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H	H		"		"
K18	"	7,20,21, 23,31	2,3,5,8, 9,13,14, 15,16,24, 26,32,35, 39,40	"	Open	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H		"		"
K19	"	2,3,5,8,9, 13,14,15, 16,21,23, 24,26,32, 35,39,40	7,20,31	"	Open	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L		"		"

See footnotes at end of table.

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TABLE IX. Functional test - Continued.

Test condition	Parameter	Connect				Ouput logic states																Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1 Q A	1 Q B	1 Q C	1 Q D	1 CAR	1 BOR	2 Q A	2 Q B	2 Q C	2 Q D	2 CAR	2 BOR	3 Q A	3 Q B	3 Q C	3 Q D	3 CAR	3 BOR	Min	Max		
K20	Functional, parallel load	2,3,5,8,9, 13,14,15, 16,21,23, 24,35,40	7,20,31	1/ 2/	26,32, 39	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L		3/		10
K21	Functional, count up	2,3,5,8,9, 13,14,15, 16,23,24, 35,40	21,26,32, 39	"	7,20, 31	H	L	L	L	L	H	H	L	L	L	H	H	H	L	L	L	H	H		"		"
K22	"	"	"	"	"	L	H	L	L	H	H	L	H	L	L	H	H	L	H	L	L	H	H		"		"
K23	"	"	"	"	"	H	H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H		"		"
K24	"	"	"	"	"	L	L	H	L	H	H	L	L	H	L	H	H	L	L	H	L	H	H		"		"
K25	"	"	"	"	"	H	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H		"		"
K26	"	"	"	"	"	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H		"		"
K27	"	"	"	"	"	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H		"		"
K28	"	"	"	"	"	L	L	L	H	H	H	L	L	L	H	H	H	L	L	L	H	H	H		"		"
K29	"	"	"	"	"	H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H		"		"
K30	"	"	"	"	"	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H	H		"		"
K31	"	"	"	"	"	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H		"		"
K32	"	"	"	"	"	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H	H		"		"
K33	"	"	"	"	"	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H		"		"
K34	"	"	"	"	"	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H		"		"
K35	"	"	"	"	"	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H		"		"
K36	"	23	2,3,5,8, 9,13,14, 15,16,21, 24,26,32, 35,39,40	"	"	L	L	L	L	H	H	L	L	L	L	H	H	L	L	L	L	H	H		"		"

See footnotes at end of table.

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TABLE IX. Functional test - Continued.

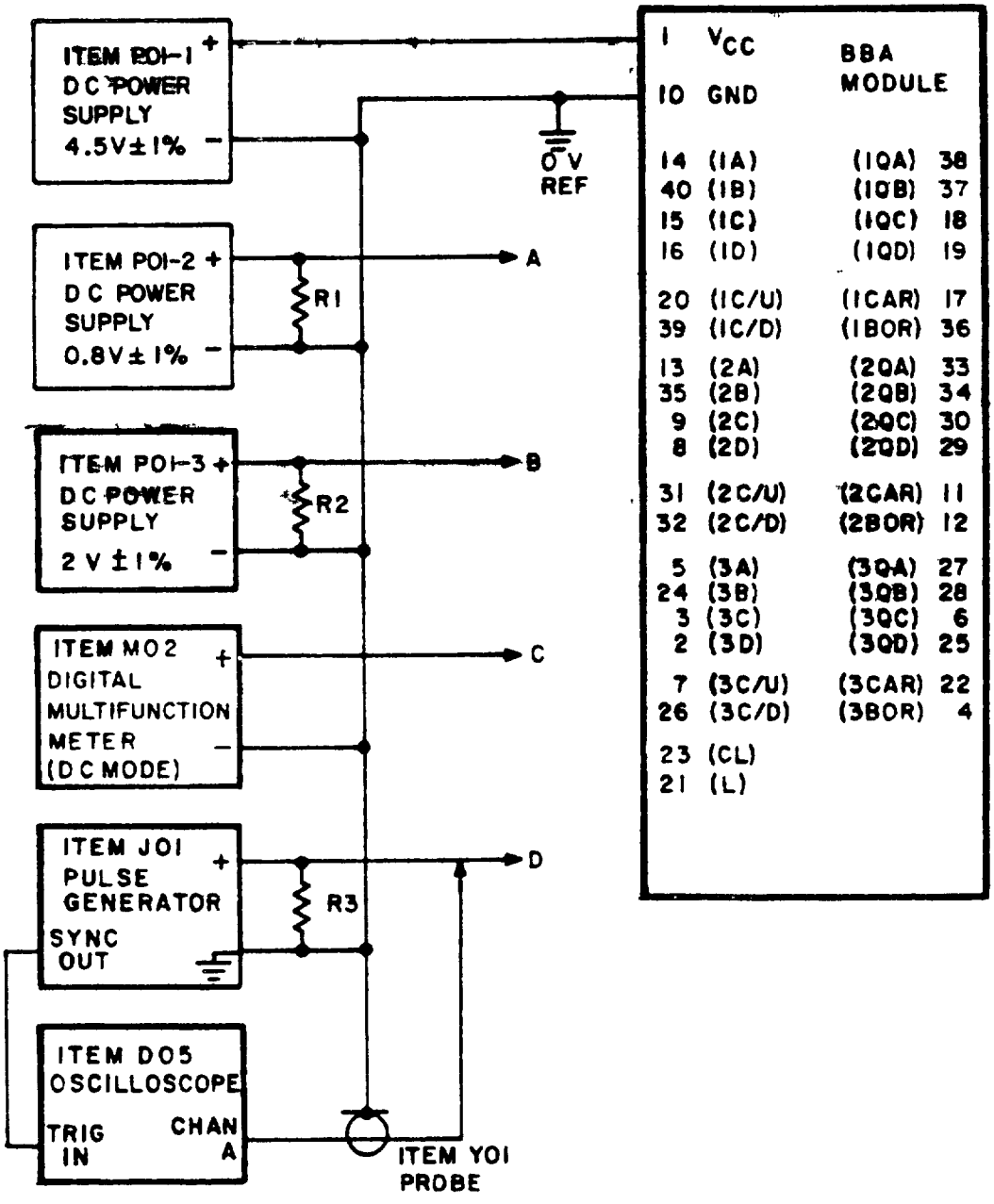
Test condition	Parameter	Connect				Ouput logic states																		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1QA	1QB	1QC	1QD	1CAR	1BQR	2QA	2QB	2QC	2QD	2CAR	2BQR	3QA	3QB	3QC	3QD	3CAR	3BQR	Min	Max		
K37	Functional, parallel load	21,23	2,3,5,7, 8,9,13, 14,15,16, 20,24,31, 35,40	1/ 2/	26,32, 39	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H		3/		10
K38	Functional, count down	23	2,3,5,7, 8,9,13, 14,15,16, 20,21,24, 31,35,40	"	"	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H		"		"
K39	"	"	"	"	"	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H	H		"		"
K40	"	"	"	"	"	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H	H		"		"
K41	"	"	"	"	"	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H	H		"		"
K42	"	"	"	"	"	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H	H		"		"
K43	"	"	"	"	"	H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H	H		"		"
K44	"	"	"	"	"	L	L	L	H	H	H	L	L	L	H	H	H	L	L	L	H	H	H		"		"
K45	"	"	"	"	"	H	H	H	L	H	H	H	H	H	L	H	H	H	H	H	L	H	H		"		"
K46	"	"	"	"	"	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H	L	H	H		"		"
K47	"	"	"	"	"	H	L	H	L	H	H	H	L	H	L	H	H	H	L	H	L	H	H		"		"
K48	"	"	"	"	"	L	L	H	L	H	H	L	L	H	L	H	H	L	L	H	L	H	H		"		"
K49	"	"	"	"	"	H	H	L	L	H	H	H	H	L	L	H	H	H	H	L	L	H	H		"		"
K50	"	"	"	"	"	L	H	L	L	H	H	L	H	L	L	H	H	L	H	L	L	H	H		"		"
K51	"	"	"	"	"	H	L	L	L	H	H	H	L	L	L	H	H	H	L	L	L	H	H		"		"
K52	"	"	"	"	"	L	L	L	L	H	L	L	L	L	L	H	L	L	L	L	L	H	L		"		"
K53	"	"	"	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H		"		"

1/ For test conditions K1-K53, verification of the output logic states in footnote 3 is obtained by connecting item MU2 (point C) to each output pin in succession after the input conditions have been accomplished. Test conditions K1-K53 are to be performed in sequence.

2/ 1QA, 1QB, 1QC, 1QD, 1CAR, 1BQR, 2QA, 2QB, 2QC, 2QD, 2CAR, 2BQR, 3QA, 3QB, 3QC, 3QD, 3CAR, and 3BQR output logic states are measured at output pins 38, 37, 18, 19, 17, 36, 33, 34, 30, 29, 11, 12, 27, 28, 6, 25, 22, and 4 respectively.

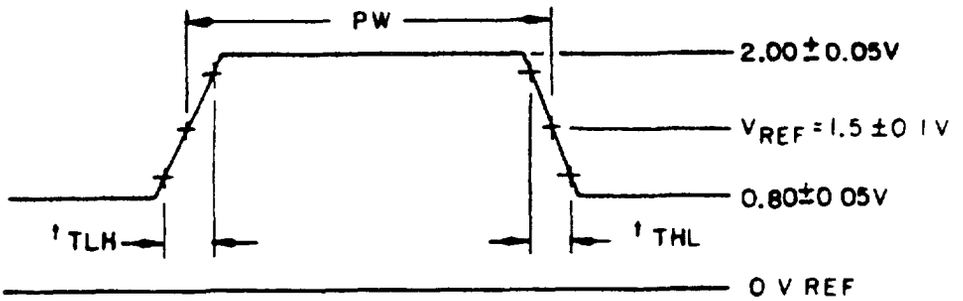
3/ Output logic states
L = Low level = 0.0 to 0.4 V.
H = High level = 2.4 to 4.5 V.

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* FIGURE 10. Output voltage (low-level and high-level) and functional test circuit for tables VIII and IX.

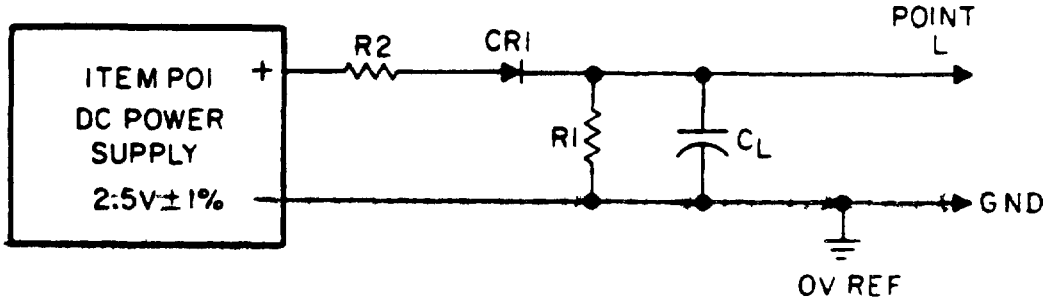
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NOTES
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. For test conditions H1-H18 and J1-J18, load each output of the circuit under test with the load as specified on figure 11.
3. For test conditions k1-K53, loading of outputs is not required.
4. For test conditions K1-K53:
   After making all input circuit connections for a test condition, use the single pulse mode on item J01 (after initial setup) and apply one pulse before making each measurement. Test conditions K1-K53 are to be performed in sequence.
5. Item J01 setup:
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 50 ±25 ns.
   Single pulse mode.

* FIGURE 10. Output voltage (low-level and high-level) and functional test circuit for tables VIII and IX - Continued.

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NOTES:
1. R1 = 6.04 kΩ ±1%, 1/8 W.
2. R2 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.4 V.
3. CR1 = 1N4150, or equivalent.
4. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions M, N, P, Q, R, S, T, U, V, W, X, Y, and Z.
5. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions BB and CC.
6. C_L is not required for test conditions H and J.

* FIGURE 11. BBA logic load circuit.

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TABLE X. Power supply current (high-level and low-level) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Power supply current, high level	21, 23	50	306	mA	12
L2	Power supply current, low level	Open	50	306	mA	12

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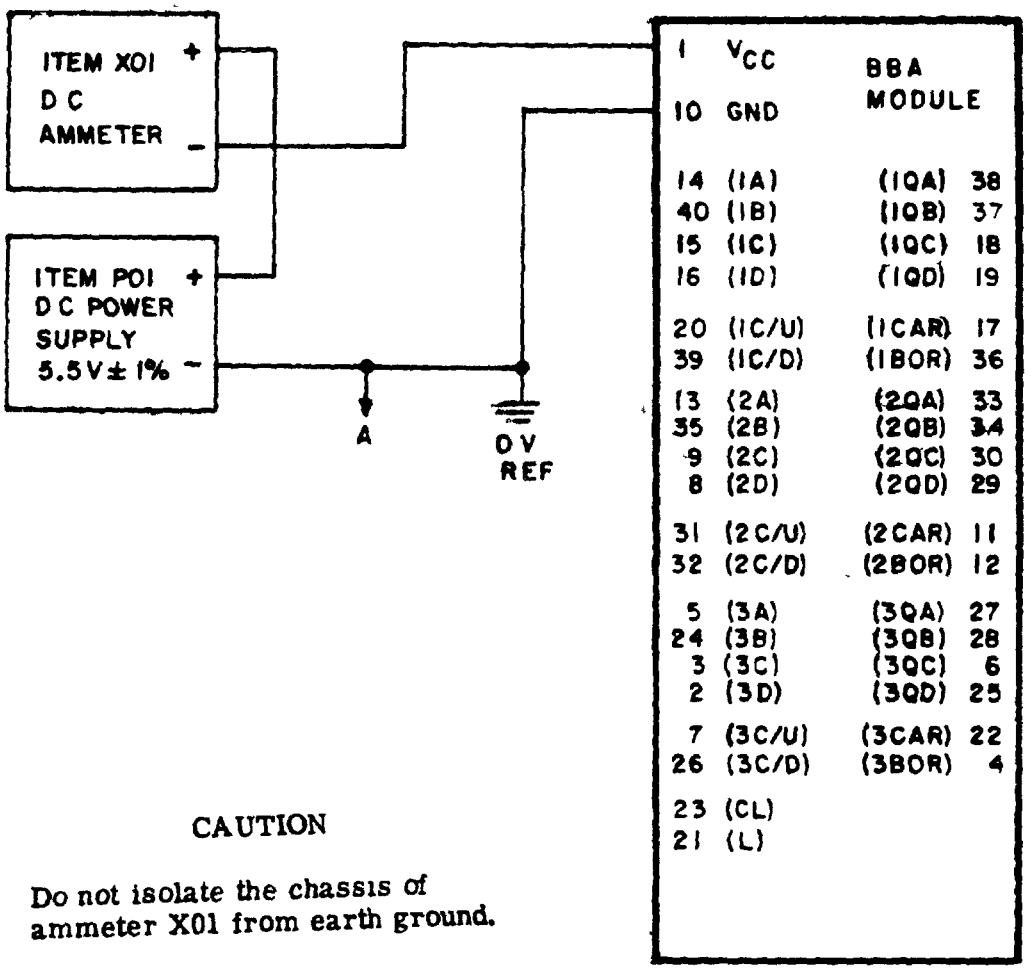


FIGURE 12. Power supply current (high-level and low-level) test circuit for table X.

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*TABLE XI. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
M1	C/U to 1CAR	Propagation delay time (t_{PHL})	14	15,16, 40	20	21	20	17	3	35	ns	13,14
M2	C/U to 2CAR	"	13	9,8,35	31	"	31	11	"	"	"	"
M3	C/U to 3CAR	"	5	3,2,24	7	"	7	22	"	"	"	"
M4	C/D to 1BOR	"	15,16, 40	14	39	"	39	36	"	"	"	"
M5	C/D to 2BOR	"	8,9,35	13	32	"	32	12	"	"	"	"
M6	C/D to 3BOR	"	2,3,24	5	26	"	26	4	"	"	"	"
N1	C/U to 1CAR	Propagation delay time (t_{PLH})	14	15,16, 40	20	"	20	17	"	"	"	"
N2	C/U to 2CAR	"	13	9,8,35	31	"	31	11	"	"	"	"
N3	C/U to 3CAR	"	5	3,2,24	7	"	7	22	"	"	"	"
N4	C/D to 1BOR	"	15,16, 40	14	39	"	39	36	"	"	"	"
N5	C/D to 2BOR	"	8,9,35	13	32	"	32	12	"	"	"	"
N6	C/D to 3BOR	"	2,3,24	5	26	"	26	4	"	"	"	"

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*TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
P1	C/U to 1QA	Propagation delay time (t_{PLH})	14,15, 16,40	Open	20	21	20	38	3	47	ns	13,15
P2	C/U to 2QA	"	8,9,13, 35	"	31	"	31	33	"	"	"	"
P3	C/U to 3QA	"	2,3,5, 24	"	7	"	7	27	"	"	"	"
P4	C/U to 1QB	"	15,16, 40	14	20	"	20	37	"	"	"	"
P5	C/U to 2QB	"	8,9,35	13	31	"	31	34	"	"	"	"
P6	C/U to 3QB	"	2,3,24	5	7	"	7	28	"	"	"	"
P7	C/U to 1QC	"	15,16	14,40	20	"	20	18	"	"	"	"
P8	C/U to 2QC	"	8,9	13,35	31	"	31	30	"	"	"	"
P9	C/U to 3QC	"	2,3	5,24	7	"	7	6	"	"	"	"
P10	C/U to 1QD	"	16	14,15, 40	20	"	20	19	"	"	"	"
P11	C/U to 2QD	"	8	9,13, 35	31	"	31	29	"	"	"	"
P12	C/U to 3QD	"	2	3,5, 24	7	"	7	25	"	"	"	"
Q1	C/U to 1QA	Propagation delay time (t_{PHL})	15,16, 40	14	20	"	20	38	"	56	"	"
Q2	C/U to 2QA	"	8,9,35	13	31	"	31	33	"	"	"	"
Q3	C/U to 3QA	"	2,3,24	5	7	"	7	27	"	"	"	"

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*TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
Q4	C/U to 1QB	Propagation delay time (t_{PHL})	15,16	14,40	20	21	20	37	3	56	ns	13,15
Q5	C/U to 2QB	"	8,9	13,35	31	"	31	34	"	"	"	"
Q6	C/U to 3QB	"	2,3	5,24	7	"	7	28	"	"	"	"
Q7	C/U to 1QC	"	16	14,15, 40	20	"	20	18	"	"	"	"
Q8	C/U to 2QC	"	8	9,13, 35	31	"	31	30	"	"	"	"
Q9	C/U to 3QC	"	2	3,5, 24	7	"	7	6	"	"	"	"
Q10	C/U to 1QD	"	Open	14,16, 15,40	20	"	20	19	"	"	"	"
Q11	C/U to 2QD	"	"	8,13, 9,35	31	"	31	29	"	"	"	"
Q12	C/U to 3QD	"	"	2,5, 3,24	7	"	7	25	"	"	"	"
R1	C/D to 1QA	Propagation delay time (t_{PLH})	14,15, 16	40	39	"	39	38	"	47	"	"
R2	C/D to 2QA	"	8,9,13	35	32	"	32	33	"	"	"	"
R3	C/D to 3QA	"	2,3,5	24	26	"	26	27	"	"	"	"
R4	C/D to 1QB	"	14,16, 40	15	39	"	39	37	"	"	"	"
R5	C/D to 2QB	"	8,13,35	9	32	"	32	34	"	"	"	"
R6	C/D to 3QB	"	2,5,24	3	26	"	26	28	"	"	"	"

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*TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
R7	C/D to 1QC	Propagation delay time (t_{PLH})	14,15, 40	16	39	21	39	18	3	47	ns	13,15
R8	C/D to 2QC	"	9,13,35	8	32	"	32	30	"	"	"	"
R9	C/D to 3QC	"	3,5,24	2	26	"	26	6	"	"	"	"
R10	C/D to 1QD	"	14,15, 16,40	Open	39	"	39	19	"	"	"	"
R11	C/D to 2QD	"	8,9,13, 35	"	32	"	32	29	"	"	"	"
R12	C/D to 3QD	"	2,3,5, 24	"	26	"	26	25	"	"	"	"
S1	C/D to 1QA	Propagation delay time (t_{PHL})	15,16, 40	14	39	"	39	38	"	56	"	"
S2	C/D to 2QA	"	8,9,35	13	32	"	32	33	"	"	"	"
S3	C/D to 3QA	"	2,3,24	5	26	"	26	27	"	"	"	"
S4	C/D to 1QB	"	14,15, 16	40	39	"	39	37	"	"	"	"
S5	C/D to 2QB	"	8,9,13	35	32	"	32	34	"	"	"	"
S6	C/D to 3QB	"	2,3,5	24	26	"	26	28	"	"	"	"
S7	C/D to 1QC	"	14,16, 40	15	39	"	39	18	"	"	"	"
S8	C/D to 2QC	"	8,13,35	9	32	"	32	30	"	"	"	"
S9	C/D to 3QC	"	2,5,24	3	26	"	26	6	"	"	"	"

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*TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
S10	C/D to 1QD	Propagation delay time (t_{PHL})	14,15, 40	16	39	21	39	19	3	56	ns	13,15
S11	C/D to 2QD	"	9,13,35	8	32	"	32	29	"	"	"	"
S12	C/D to 3QD	"	3,5,24	2	26	"	26	25	"	"	"	"

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test.

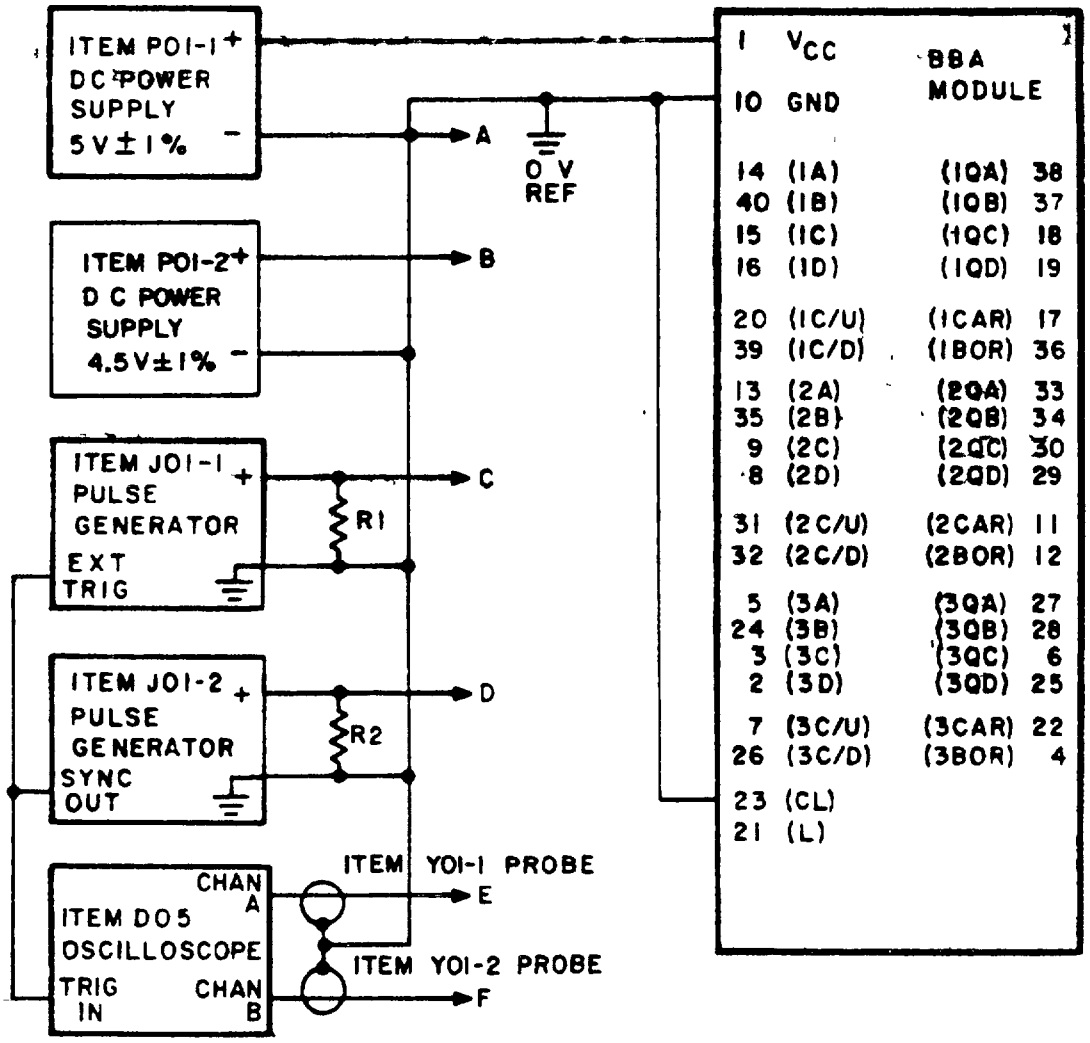
Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
T1	1CAR	Transition time (t_{TLH})	14	15,16, 40	20	21	20	17	10	45	ns	13,14
T2	2CAR	"	13	9,8,35	31	"	31	11	"	"	"	"
T3	3CAR	"	5	3,2,24	7	"	7	22	"	"	"	"
T4	1BOR	"	15,16, 40	14	39	"	39	36	"	"	"	"
T5	2BOR	"	8,9,35	13	32	"	32	12	"	"	"	"
T6	3BOR	"	2,3,24	5	26	"	26	4	"	"	"	"
T7	1QA	"	14,15, 16,40	Open	20	"	20	38	"	"	"	13,15
T8	2QA	"	8,9,13, 35	"	31	"	31	33	"	"	"	"
T9	3QA	"	2,3,5, 24	"	7	"	7	27	"	"	"	"
T10	1QB	"	15,16, 40	14	20	"	20	37	"	"	"	"
T11	2QB	"	8,9,35	13	31	"	31	34	"	"	"	"
T12	3QB	"	2,3,24	5	7	"	7	28	"	"	"	"
T13	1QC	"	15,16	14,40	20	"	20	18	"	"	"	"
T14	2QC	"	8,9	13,35	31	"	31	30	"	"	"	"
T15	3QC	"	2,3	5,24	7	"	7	6	"	"	"	"
T16	1QD	"	16	14,15, 40	20	"	20	19	"	"	"	"
T17	2QD	"	8	9,13, 35	31	"	31	29	"	"	"	"
T18	3QD	"	2	3,5,24	7	"	7	25	"	"	"	"
U1	1CAR	Transition time (t_{THL})	14	15,16, 40	20	"	20	17	5	20	"	13,14

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
U2	2CAR	Transition time (t_{THL})	13	8,9,35	31	21	31	11	5	20	ns	13,14
U3	3CAR	"	5	2,3,24	7	"	7	22	"	"	"	"
U4	1BOR	"	15,16, 40	14	39	"	39	36	"	"	"	"
U5	2BOR	"	8,9,35	13	32	"	32	12	"	"	"	"
U6	3BOR	"	2,3,24	5	26	"	26	4	"	"	"	"
U7	1QA	"	15,16, 40	14	20	"	20	38	"	"	"	13,15
U8	2QA	"	8,9,35	13	31	"	31	33	"	"	"	"
U9	3QA	"	2,3,24	5	7	"	7	27	"	"	"	"
U10	1QB	"	15,16	14,40	20	"	20	37	"	"	"	"
U11	2QB	"	8,9	13,35	31	"	31	34	"	"	"	"
U12	3QB	"	2,3	5,24	7	"	7	28	"	"	"	"
U13	1QC	"	16	14,15, 40	20	"	20	18	"	"	"	"
U14	2QC	"	8	9,13, 35	31	"	31	30	"	"	"	"
U15	3QC	"	2	3,5,24	7	"	7	6	"	"	"	"
U16	1QD	"	Open	14,16, 15,40	20	"	20	19	"	"	"	"
U17	2QD	"	"	8,13, 9,35	31	"	31	29	"	"	"	"
U18	3QD	"	"	2,5,3, 24	7	"	7	25	"	"	"	"

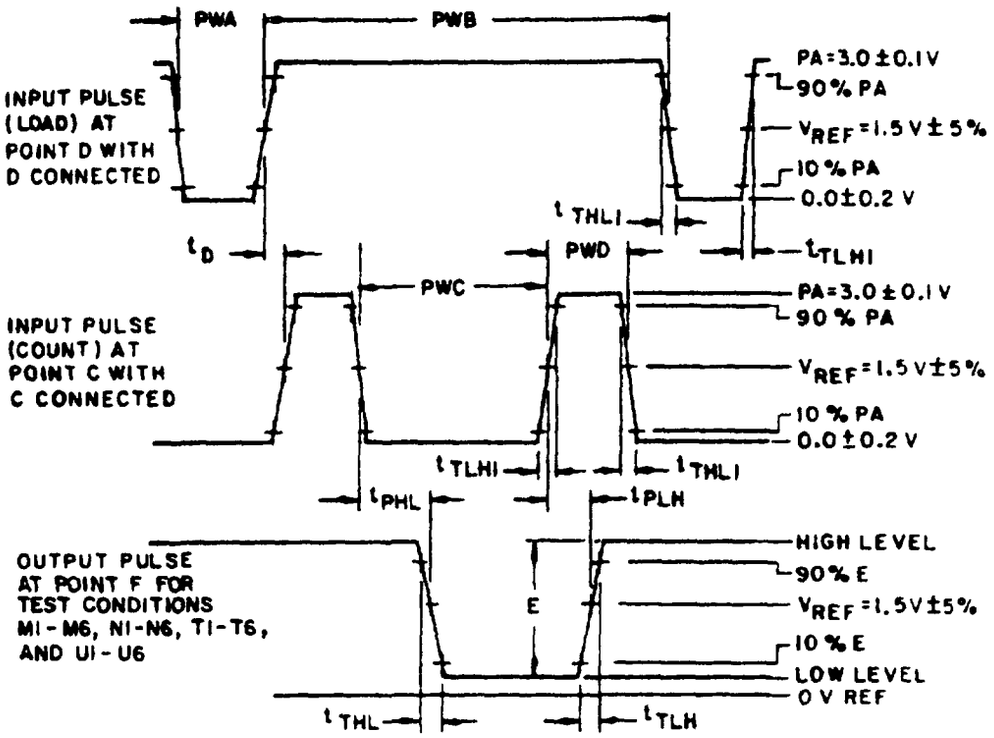
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 11.

* FIGURE 13. Propagation delay time (t_{PHL} and t_{PLH}), propagation delay time (t_{PLH} and t_{PHL}) and transition time (t_{TLH} and t_{THL}) test circuit for tables XI, XII, and XIII.

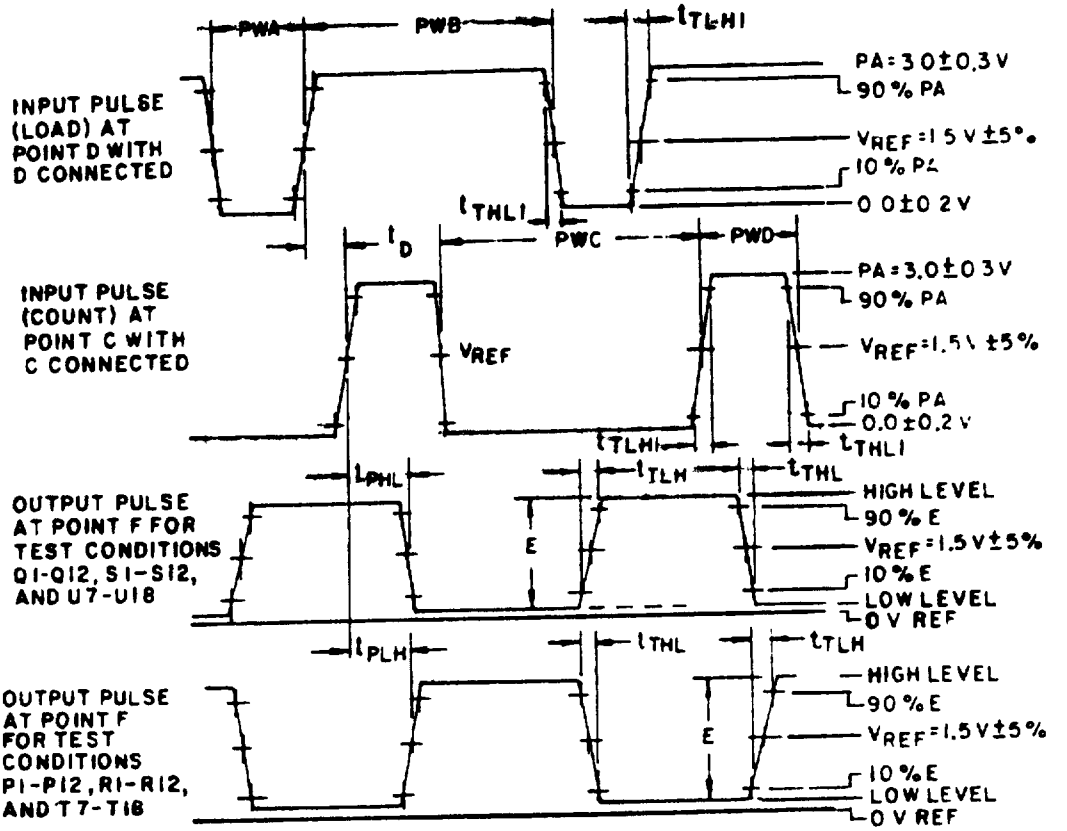
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NOTES:
1. Input pulse parameters:
   PWA = 100 ±10 ns.
   PWB = 900 ±90 ns.
   PWC = 400 ±40 ns.
   PWD = 100 ±10 ns.
   t_D = 100 ±10 ns.
   t_{TLH1} = t_{THL1} = 10 ±2 ns.
2. Verify that the output low level is $\leq$ 0.4 V and the output high level is $\geq$ 2.4 V.
3. Pulse at point C is obtained by placing item J01-1 in the double pulse mode.
4. All voltage levels are referenced to ground.

* FIGURE 14. Propagation delay time (t_{PHL} and t_{PLH}) and transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XI and XIII.

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NOTES:

1. Input pulse parameters:
   PWA = 100 ±10 ns.
   PWB = 900 ±90 ns.
   PWC = 900 ±90 ns.
   PWD = 100 ±10 ns.
   t_D = 100 ±10 ns.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Verify that the output low level is $\leq$ 0.4 V and the output high level is $\geq$ 2.4 V.
3. All voltage levels are referenced to ground.

* FIGURE 15. Propagation delay time (t_{PLH} and t_{PHL}) and transition time (t_{TLH} and t_{THL}) test circuit waveforms for tables XII and XIII.

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*TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) (minimum load hold time check and minimum clear pulse width check) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V1	L to 1QA(t_{PLH}) Propagation delay time	15,16, 23,40	Open	14	21	38	3	49	ns	16,17
W1	L to 1QA(t_{PHL}) "	15,16, 23,40	"	14	"	38	"	"	"	"
V2	L to 2QA(t_{PLH}) "	8,9,23, 35	"	13	"	33	"	"	"	"
W2	L to 2QA(t_{PHL}) "	8,9,23, 35	"	13	"	33	"	"	"	"
V3	L to 3QA(t_{PLH}) "	2,3,23, 24	"	5	"	27	"	"	"	"
W3	L to 3QA(t_{PHL}) "	2,3,23, 24	"	5	"	27	"	"	"	"
V4	L to 1QB(t_{PLH}) "	14,15, 16,23	"	40	"	37	"	"	"	"
W4	L to 1QB(t_{PHL}) "	14,15, 16,23	"	40	"	37	"	"	"	"
V5	L to 2QB(t_{PLH}) "	8,9,13, 23	"	35	"	34	"	"	"	"
W5	L to 2QB(t_{PHL}) "	8,9,13, 23	"	35	"	34	"	"	"	"
V6	L to 3QB(t_{PLH}) "	2,3,5, 23	"	24	"	28	"	"	"	"
W6	L to 3QB(t_{PHL}) "	2,3,5, 23	"	24	"	28	"	"	"	"
V7	L to 1QC(t_{PLH}) "	14,16, 23,40	"	15	"	18	"	"	"	"
W7	L to 1QC(t_{PHL}) "	14,16, 23,40	"	15	"	18	"	"	"	"
V8	L to 2QC(t_{PLH}) "	8,13, 23,35	"	9	"	30	"	"	"	"

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*TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) (minimum load hold time check and minimum clear pulse width check) test - Continued.

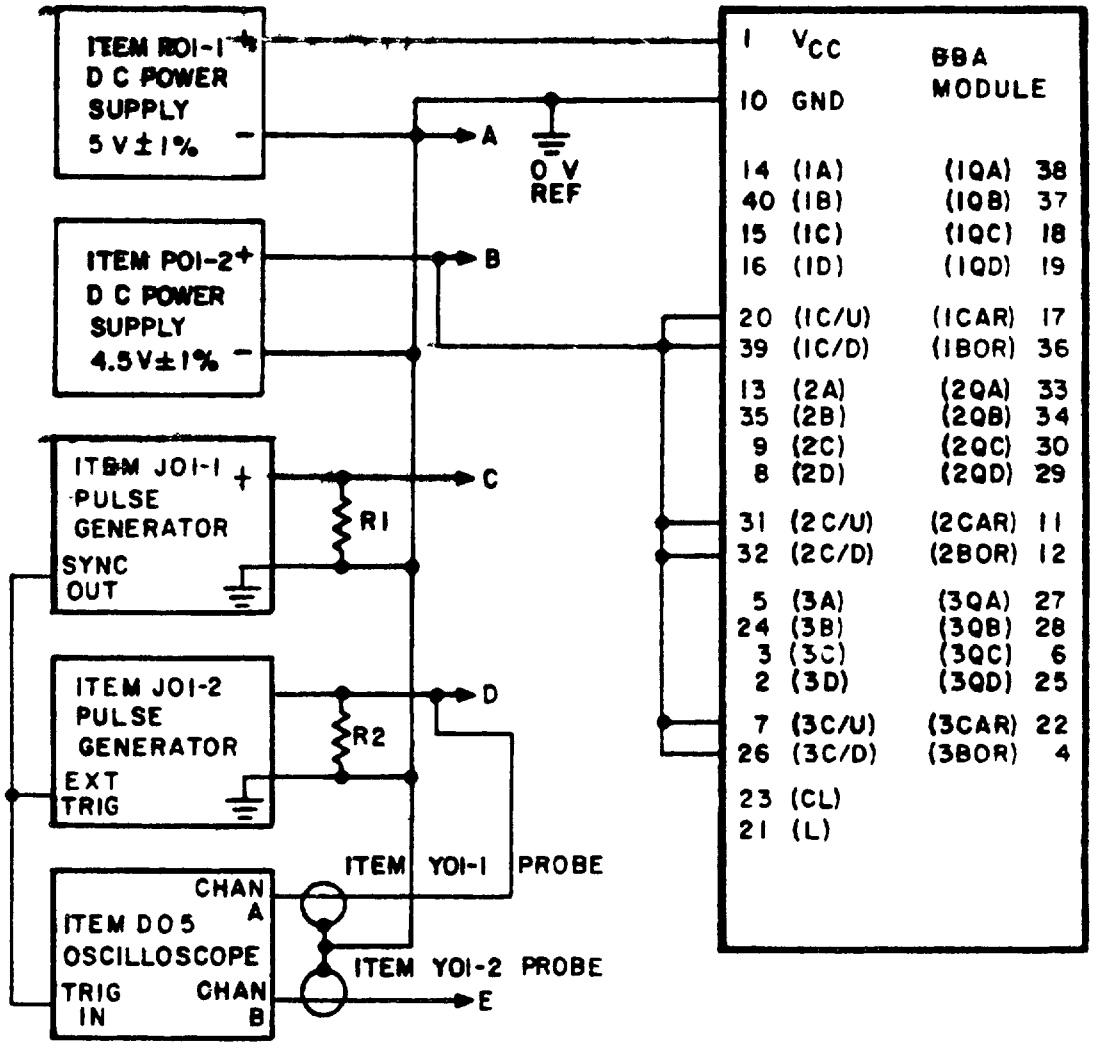
Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W8	L to 2QC(t_{PHL}) Propagation delay time	8,13, 23,35	Open	9	21	30	3	49	ns	16,17
V9	L to 3QC(t_{PLH}) "	2,5, 23,24	"	3	"	6	"	"	"	"
W9	L to 3QC(t_{PHL}) "	2,5, 23,24	"	3	"	6	"	"	"	"
V10	L to 1QD(t_{PLH}) "	14,15, 23,40	"	16	"	19	"	"	"	"
W10	L to 1QD(t_{PHL}) "	14,15, 23,40	"	16	"	19	"	"	"	"
V11	L to 2QD(t_{PLH}) "	9,13, 23,35	"	8	"	29	"	"	"	"
W11	L to 2QD(t_{PHL}) "	9,13, 23,35	"	8	"	29	"	"	"	"
V12	L to 3QD(t_{PLH}) "	3,5, 23,24	"	2	"	25	"	"	"	"
W12	L to 3QD(t_{PHL}) "	3,5, 23,24	"	2	"	25	"	"	"	"
X1	CL to 1QA(t_{PHL}) "	15,16, 40	14	21	23	38	"	44	"	16,18
X2	CL to 2QA(t_{PHL}) "	8,9,35	13	"	"	33	"	"	"	"
X3	CL to 3QA(t_{PHL}) "	2,3,24	5	"	"	27	"	"	"	"
X4	CL to 1QB(t_{PHL}) "	14,15, 16	40	"	"	37	"	"	"	"
X5	CL to 2QB(t_{PHL}) "	8,9,13	35	"	"	34	"	"	"	"
X6	CL to 3QB(t_{PHL}) "	2,3,5	24	"	"	28	"	"	"	"

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*TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) (minimum load hold time check and minimum clear pulse width check) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
X7	CL to 1QC(t_{PHL}) Propagation delay time	14,16, 40	15	21	23	18	3	44	ns	16,18
X8	CL to 2QC(t_{PHL}) "	8,13, 35	9	"	"	30	"	"	"	"
X9	CL to 3QC(t_{PHL}) "	2,5, 24	3	"	"	6	"	"	"	"
X10	CL to 1QD(t_{PHL}) "	14,15, 40	16	"	"	19	"	"	"	"
X11	CL to 2QD(t_{PHL}) "	9,13, 35	8	"	"	29	"	"	"	"
X12	CL to 3QD(t_{PHL}) "	3,5, 24	2	"	"	25	"	"	"	"

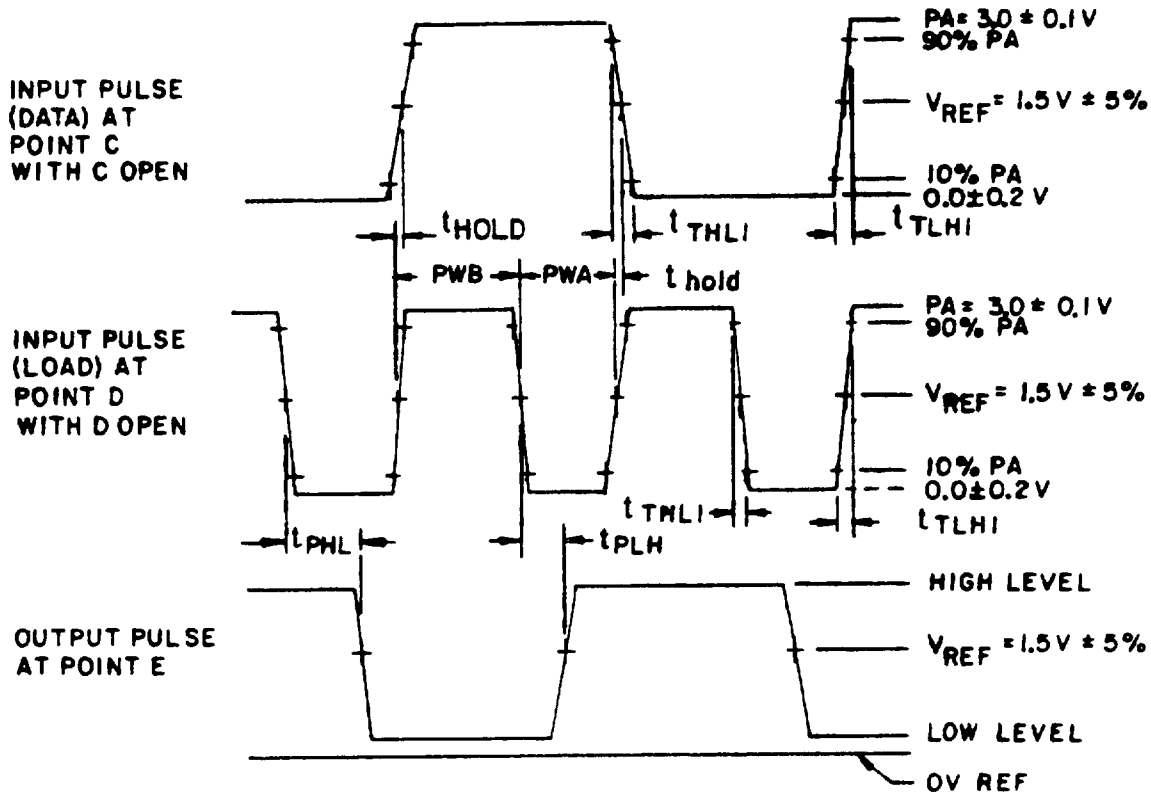
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 11.

* FIGURE 16. Propagation delay time (t_{PLH} and t_{PHL}) (minimum load hold time check and minimum clear pulse width check) test circuit for table XIV.

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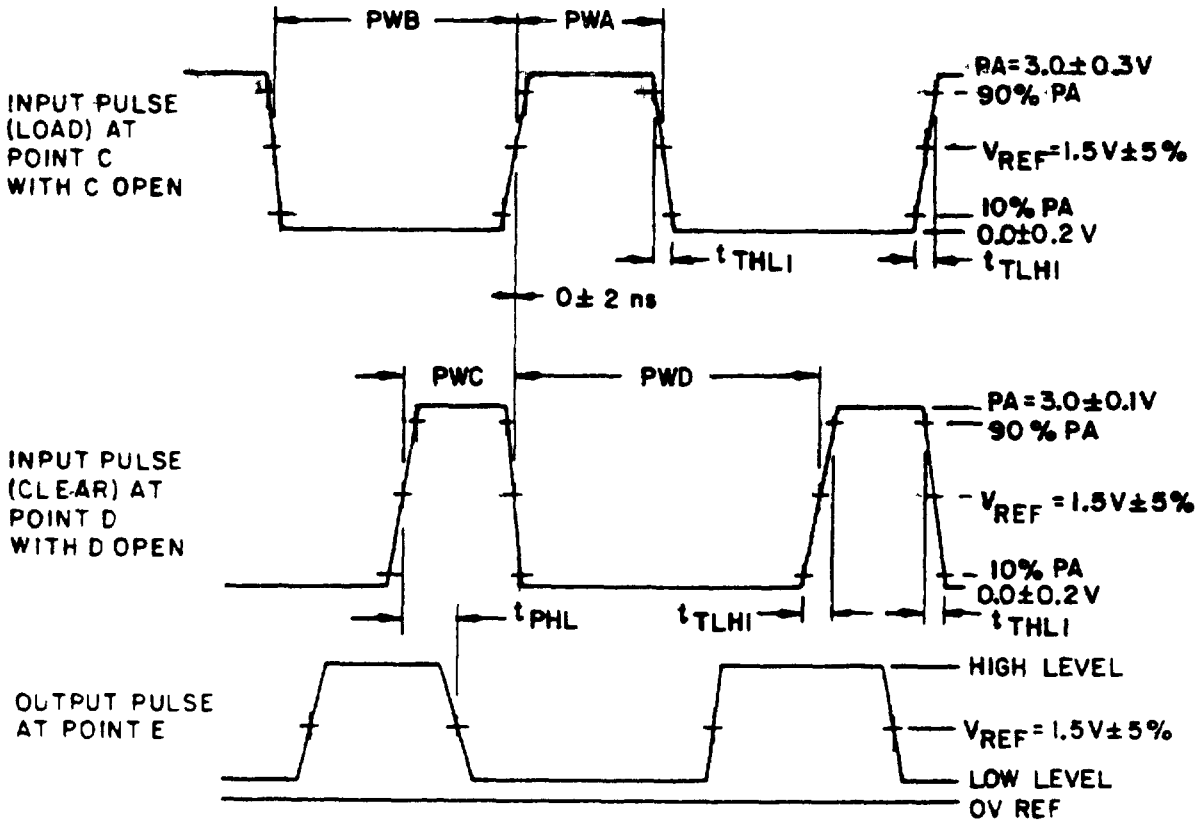


NOTES:

1. Input pulse parameters:
   PWA = $35\ ^{+2}_{-0}$ ns.
   PWB = 65 ±5 ns.
   $t_{HOLD} = 5\ ^{+2}_{-0}$ ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 2$ ns.
2. Verify that the output low level is $\leq$ 0.4 V and the output high level is $\geq$ 2.4 V.
3. All voltage levels are referenced to ground.

* FIGURE 17. Propagation delay time (t_{PLH} and t_{PHL}) (minimum load hold time check) test circuit waveforms for table XIV.

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NOTES:
1. Input pulse parameters:
   PWA = 40 ±4 ns.
   PWB = 60 ±6 ns.
   PWC = 25 $^{+2}_{-0}$ ns (clear pulse width).
   PWD = 80 ±8 ns.
   $t_{TLHI} = t_{THLI}$ = 6 ±2 ns.
2. Verify that the output low level is $\leq$ 0.4 V and the output high level is $\geq$ 2.4 V.
3. All voltage levels are referenced to ground.

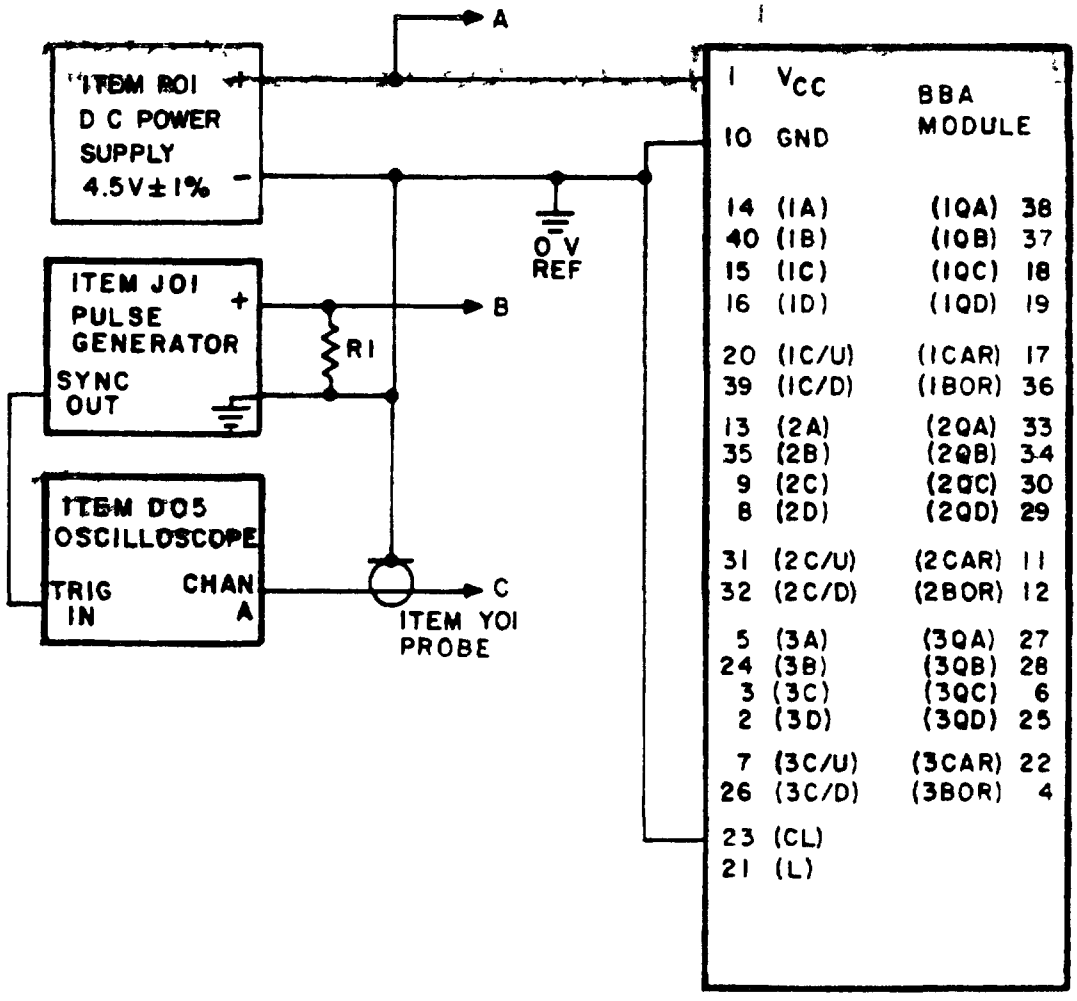
* FIGURE 18. Propagation delay time (t_{PLH}) (minimum clear pulse width check) test circuit waveforms for table XIV.

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TABLE XV. Maximum counting frequency check (minimum count pulse width check) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Y1	Maximum counting frequency check (minimum count pulse width check), carry period, circuit 1	39, 21	20	17	1,070	1,140	ns	19,20
Y2	Maximum counting frequency check (minimum count pulse width check), carry period, circuit 2	32, 21	31	11	"	"	"	"
Y3	Maximum counting frequency check (minimum count pulse width check), carry period, circuit 3	26, 21	7	22	"	"	"	"
Y4	Maximum counting frequency check (minimum count pulse width check), borrow period, circuit 1	20, 21	39	36	"	"	"	"
Y5	Maximum counting frequency check (minimum count pulse width check), borrow period, circuit 2	31, 21	32	12	"	"	"	"
Y6	Maximum counting frequency check (minimum count pulse width check), borrow period, circuit 3	7, 21	26	4	"	"	"	"

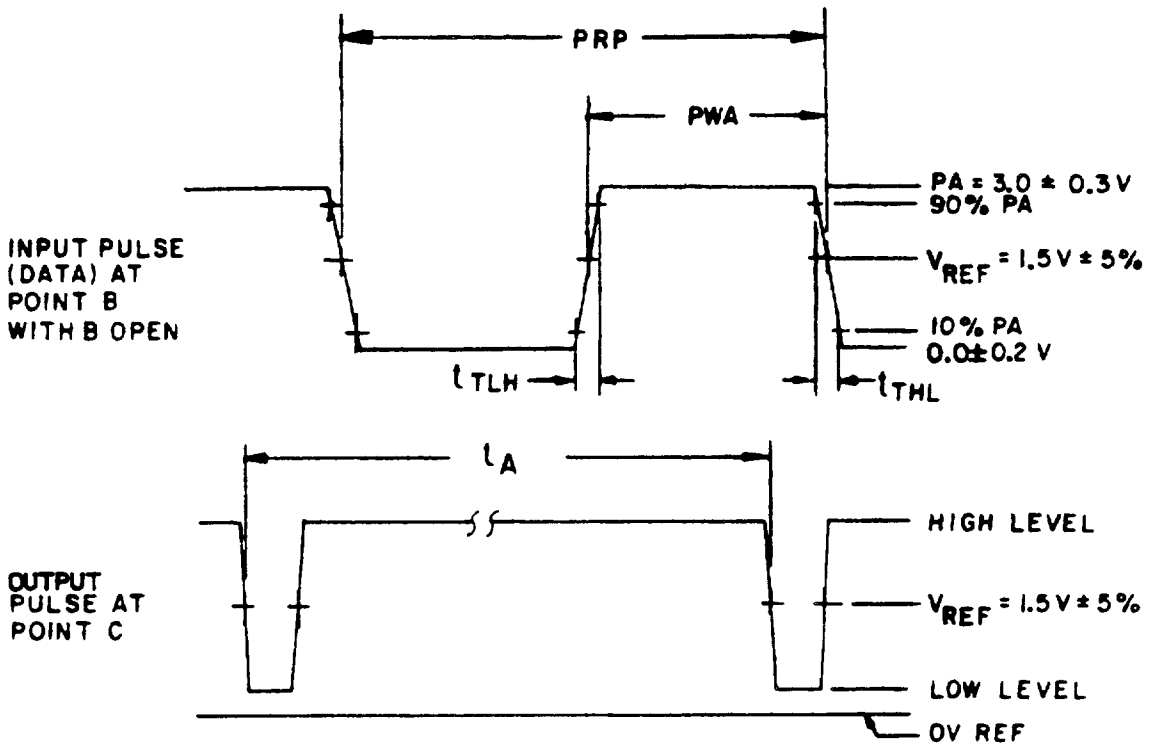
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 11.

FIGURE 19. Maximum counting frequency check (minimum count pulse width check) test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   PWA = 40 $^{+2}_{-0}$ ns.
   PRP = 69 ±1 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
2. Verify that the output low level is ≤ 0.4 V and the output high level is ≥ 2.4 V.
3. All voltage levels are referenced to ground.
4. t_A = CARRY or BORROW period.

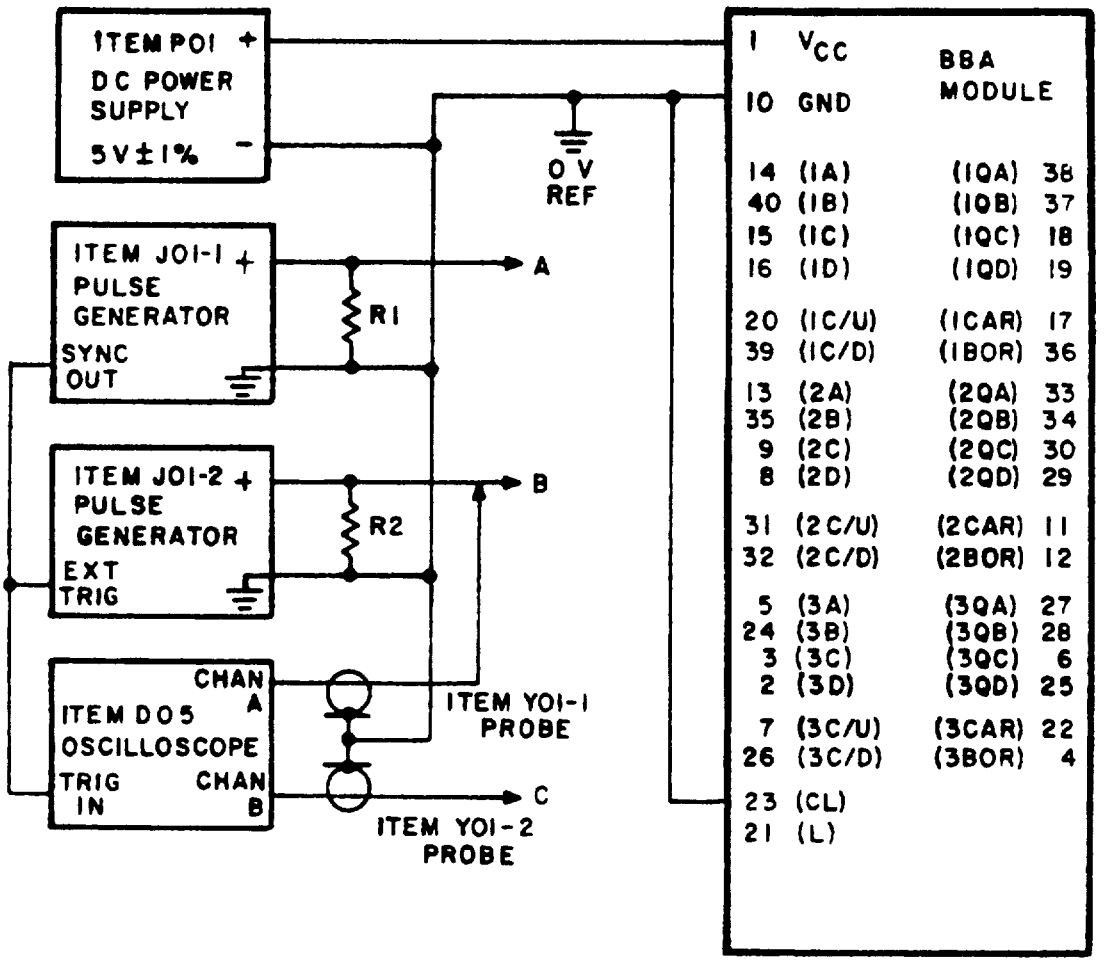
* FIGURE 20. Maximum counting frequency check (minimum count pulse width check) test circuit waveforms for table XV.

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*TABLE XVI. Setup time (data to load) and minimum load pulse width check test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Z1	1QA	Setup time (data to load) and minimum load pulse width check	14	21	38	186	214	ns	21,22
Z2	2QA	"	13	"	33	"	"	"	"
Z3	3QA	"	5	"	27	"	"	"	"
Z4	1QB	"	40	"	37	"	"	"	"
Z5	2QB	"	35	"	34	"	"	"	"
Z6	3QB	"	24	"	28	"	"	"	"
Z7	1QC	"	15	"	18	"	"	"	"
Z8	2QC	"	9	"	30	"	"	"	"
Z9	3QC	"	3	"	6	"	"	"	"
Z10	1QD	"	16	"	19	"	"	"	"
Z11	2QD	"	8	"	29	"	"	"	"
Z12	3QD	"	2	"	25	"	"	"	"

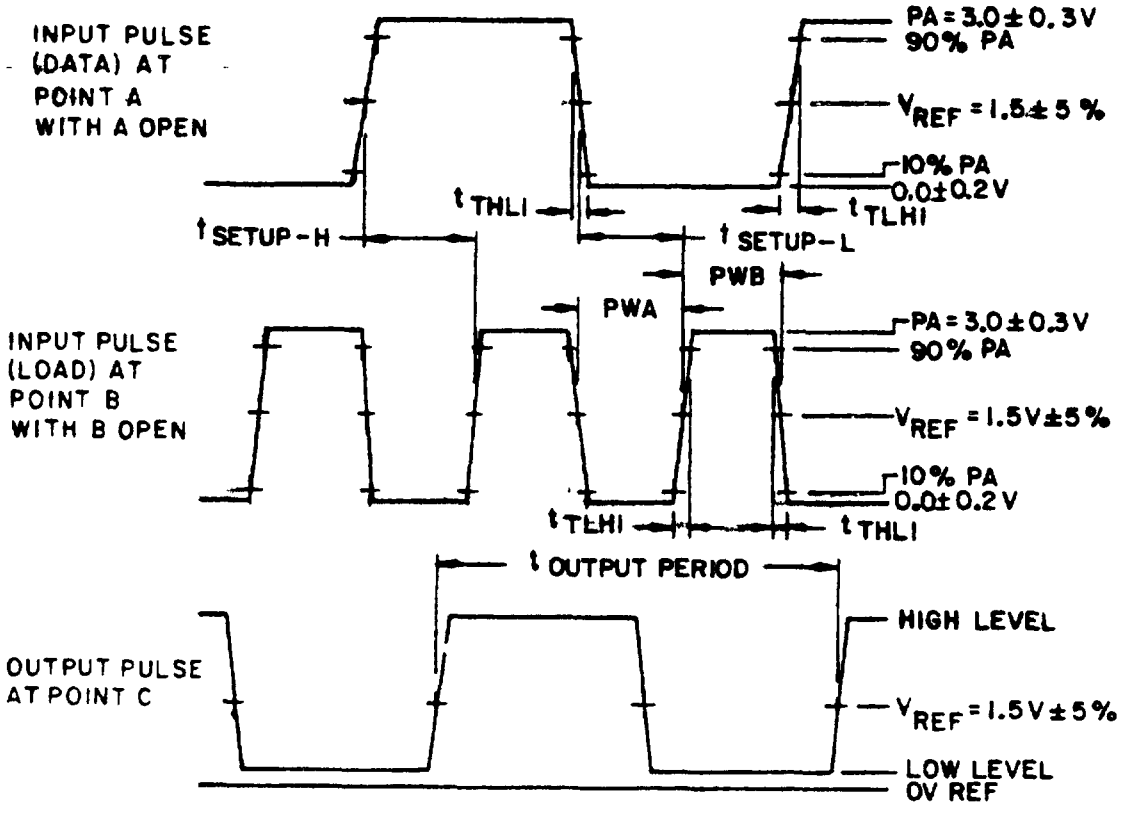
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with the load as specified on figure 11.

FIGURE 21. Setup time (data to load) and minimum load pulse width check test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   PWA = $35\ ^{+2}_{-0}$ ns.
   PWB = 65 ±5 ns.
   $t_{SETUP-H} = t_{SETUP-L} = 35\ ^{+2}_{-0}$ ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 2$ ns.
2. Verify that the output low level is $\leq$ 0.4 V and the output high level is $\geq$ 2.4 V.
3. All voltage levels are referenced to ground.

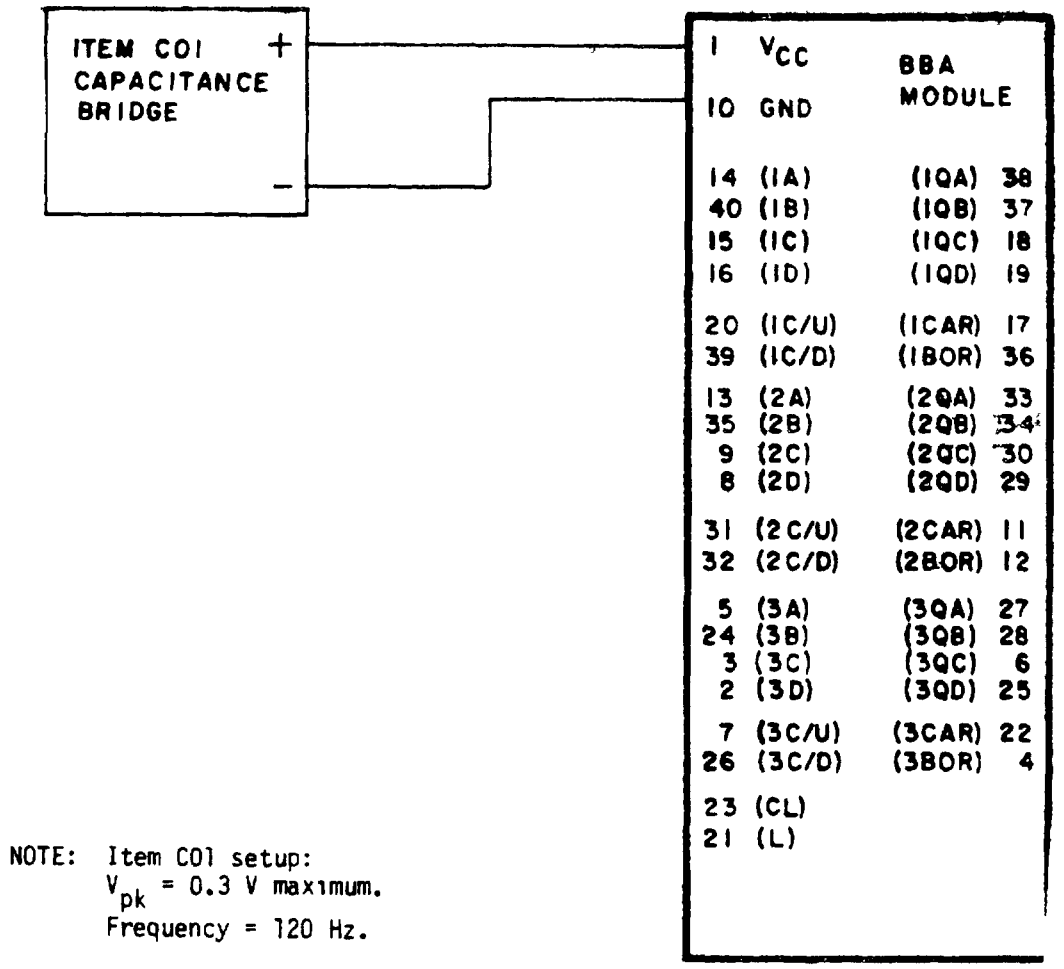
* FIGURE 22. Setup time (data to load) and minimum load pulse width check test circuit waveforms for table XVI.

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TABLE XVII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
AA	Power supply filter capacitance	1.6	3.0	µF	23

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NOTE: Item C01 setup:
V_{pk} = 0.3 V maximum.
Frequency = 120 Hz.

FIGURE 23. Power supply filter capacitance test circuit for table XVII.

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TABLE XVIII. Output crosstalk (high-level, low-level and duration) test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB1	1QA	Output cross-talk, high level	14	Open	21	2,3,5,8,9,13,15,16,24,35,40	38	0	400	mV	24,25
BB2	1QB	"	40	"	"	2,3,5,8,9,13,14,15,16,24,35	37	"	"	"	"
BB3	1QC	"	15	"	"	2,3,5,8,9,13,14,16,24,35,40	18	"	"	"	"
BB4	1QD	"	16	"	"	2,3,5,8,9,13,14,15,24,35,40	19	"	"	"	"
BB5	2QA	"	13	"	"	2,3,5,8,9,14,15,16,24,35,40	33	"	"	"	"
BB6	2QB	"	35	"	"	2,3,5,8,9,13,14,15,16,24,40	34	"	"	"	"
BB7	2QC	"	9	"	"	2,3,5,8,13,14,15,16,24,35,40	30	"	"	"	"
BB8	2QD	"	8	"	"	2,3,5,9,13,14,15,16,24,35,40	29	"	"	"	"
BB9	3QA	"	5	"	"	2,3,8,9,13,14,15,16,24,35,40	27	"	"	"	"

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TABLE XVIII. Output crosstalk (high-level, low-level and duration) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
BB10	3QB	Output crosstalk, high level	24	Open	21	2,3,5,8,9,13,14,15,16,35,40	28	0	400	mV	24,25
BB11	3QC	"	3	"	"	2,5,8,9,13,14,15,16,24,35,40	6	"	"	"	"
BB12	3QD	"	2	"	"	3,5,8,9,13,14,15,16,24,35,40	25	"	"	"	"
CC1	1QA	Output crosstalk, low level	Open	14	"	2,3,5,8,9,13,15,16,24,35,40	38	"	1	V	"
CC2	1QB	"	"	40	"	2,3,5,8,9,13,14,15,16,24,35	37	"	"	"	"
CC3	1QC	"	"	15	"	2,3,5,8,9,13,14,16,24,35,40	18	"	"	"	"
CC4	1QD	"	"	16	"	2,3,5,8,9,13,14,15,24,35,40	19	"	"	"	"
CC5	2QA	"	"	13	"	2,3,5,8,9,14,15,16,24,35,40	33	"	"	"	"
CC6	2QB	"	"	35	"	2,3,5,8,9,13,14,15,16,24,40	34	"	"	"	"
CC7	2QC	"	"	9	"	2,3,5,8,13,14,15,16,24,35,40	30	"	"	"	"
CC8	2QD	"	"	8	"	2,3,5,9,13,14,15,16,24,35,40	29	"	"	"	"

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TABLE XVIII. Output crosstalk (high-level, low-level and duration) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC9	3QA	Output cross-talk, low level	Open	5	21	2,3,8,9,13, 14,15,16,24, 35,40	27	0	1	V	24,25
CC10	3QB	"	"	24	"	2,3,5,8,9, 13,14,15,16, 35,40	28	"	"	"	"
CC11	3QC	"	"	3	"	2,5,8,9,13, 14,15,16,24, 35,40	6	"	"	"	"
CC12	3QD	"	"	2	"	3,5,8,9,13, 14,15,16,24, 35,40	25	"	"	"	"
CC13	1QA	Output cross-talk, duration	"	14	"	2,3,5,8,9,13, 15,16,24,35, 40	38	"	400	mV	24,26
CC14	1QB	"	"	40	"	2,3,5,8,9,13, 14,15,16,24, 35	37	"	"	"	"
CC15	1QC	"	"	15	"	2,3,5,8,9,13, 14,16,24,35, 40	18	"	"	"	"
CC16	1QD	"	"	16	"	2,3,5,8,9,13, 14,15,24,35, 40	19	"	"	"	"
CC17	2QA	"	"	13	"	2,3,5,8,9,14, 15,16,24,35, 40	33	"	"	"	"
CC18	2QB	"	"	35	"	2,3,5,8,9,13, 14,15,16,24, 40	34	"	"	"	"
CC19	2QC	"	"	9	"	2,3,5,8,13, 14,15,16,24, 35,40	30	"	"	"	"
CC20	2QD	"	"	8	"	2,3,5,9,13, 14,15,16,24, 35,40	29	"	"	"	"

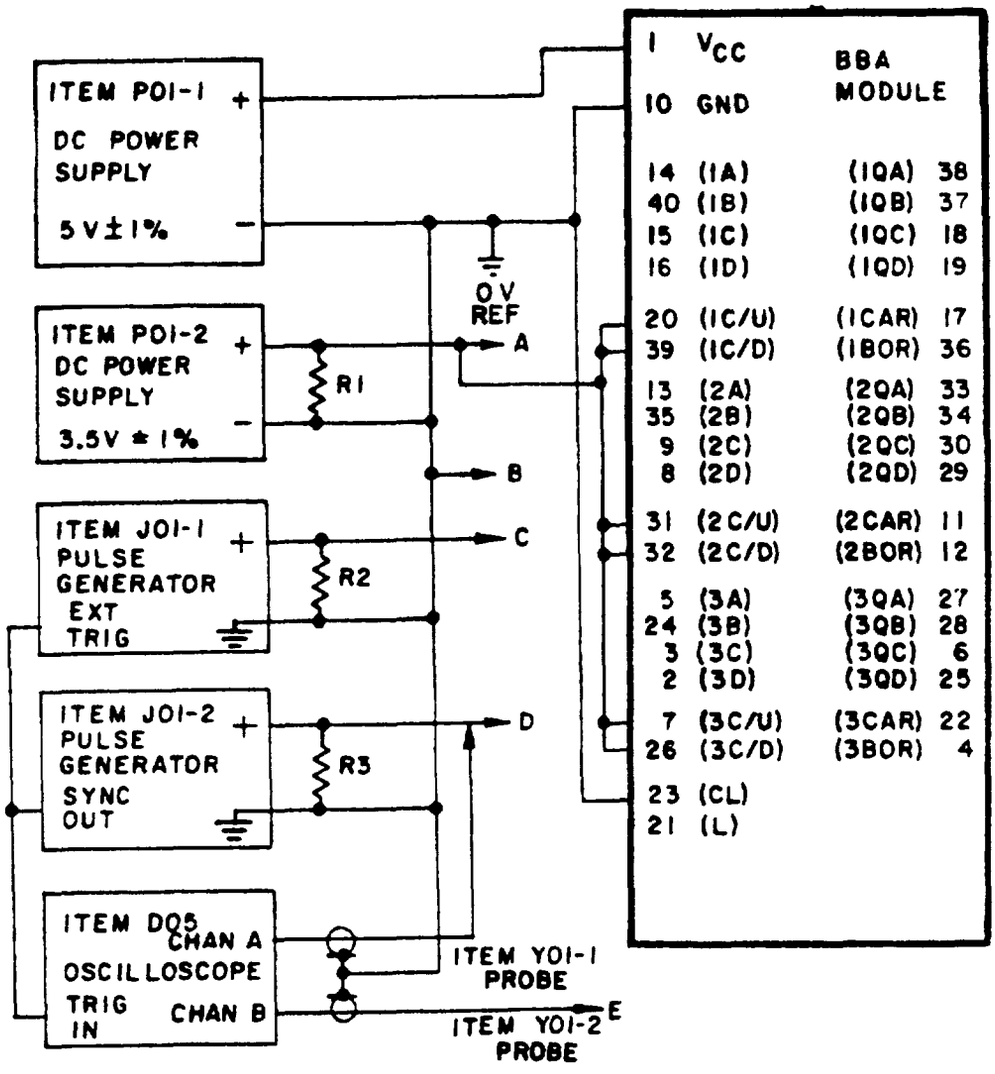
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TABLE XVIII. Output crosstalk (high-level, low-level and duration) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC21	3QA	Output cross-talk, duration	Open	5	21	2,3,8,9,13, 14,15,16,24, 35,40	27	0	400	mV	24,26
CC22	3QB	"	"	24	"	2,3,5,8,9,13, 14,15,16,35, 40	28	"	"	"	"
CC23	3QC	"	"	3	"	2,5,8,9,13, 14,15,16,24, 35,40	6	"	"	"	"
CC24	3QD	"	"	2	"	3,5,8,9,13, 14,15,16,24, 35,40	25	"	"	"	"

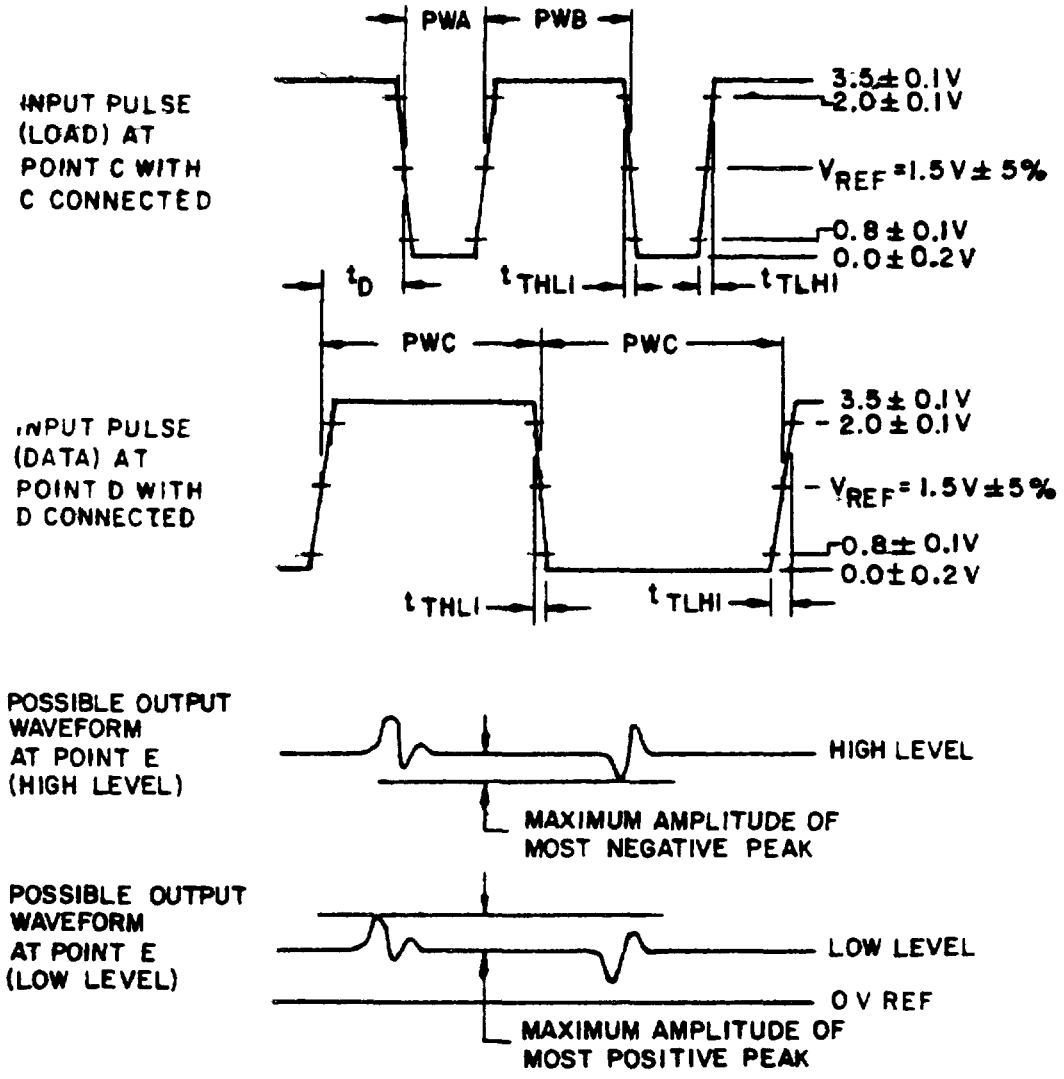
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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 11.

FIGURE 24. Output crosstalk (high-level, low-level,and duration) test circuit for table XVIII.

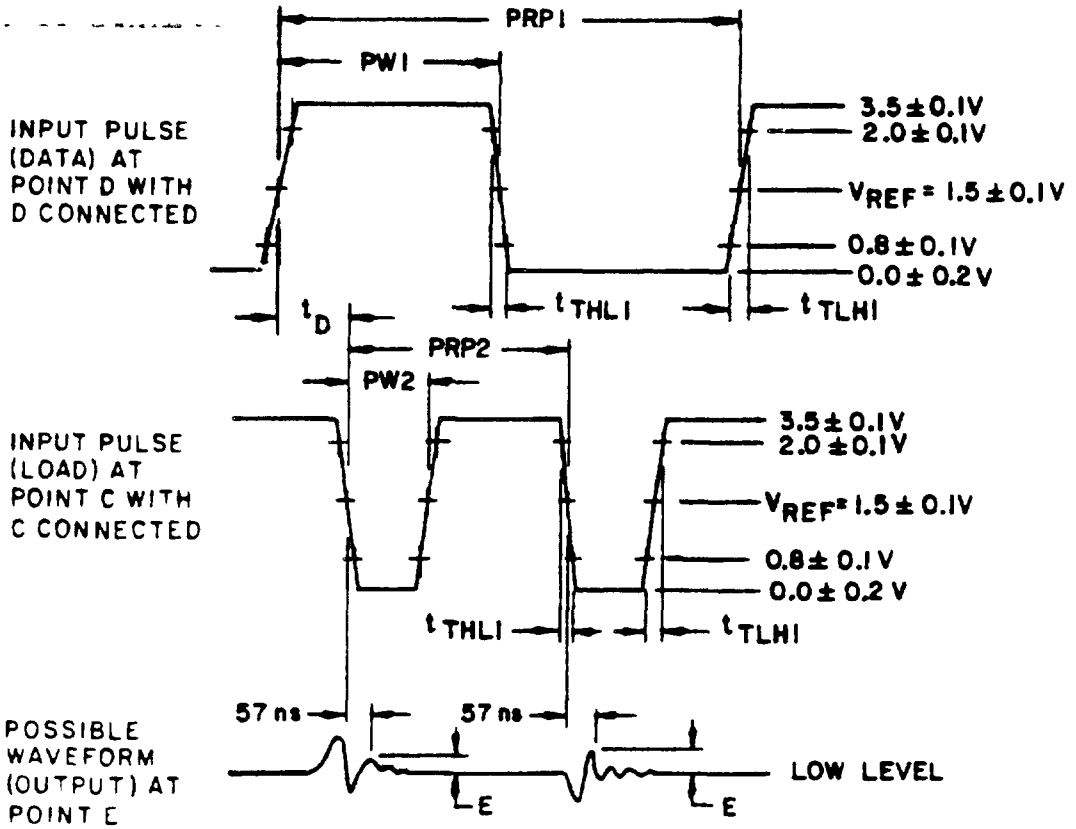
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NOTE: Input pulse parameters:
PWA = 75 ±5 ns.
PWB = 175 ±10 ns.
PWC = 250 ±25 ns.
$t_{TLH1} = t_{THL1}$ = 3.5 ±1.5 ns (verified at each input pin).
t_D = 85 ±5 ns.

* FIGURE 25. Output crosstalk (high-level and low-level) test circuit waveforms for table XVIII.

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NOTES:

1. Input pulse parameters:
   PRP1 = 500 ±50 ns.
   PW1 = 250 ±25 ns.
   PRP2 = 250 ±25 ns.
   PW2 = 75 ±5 ns.
   t_D = 85 ±5 ns.
   $t_{TLH1} = t_{THL1}$ = 3.5 ±1.5 ns (verified at each input pin).
2. E is the maximum amplitude of the most positive peak measured after a settle time of 57 ns has elapsed after any negative transition of the load pulse.

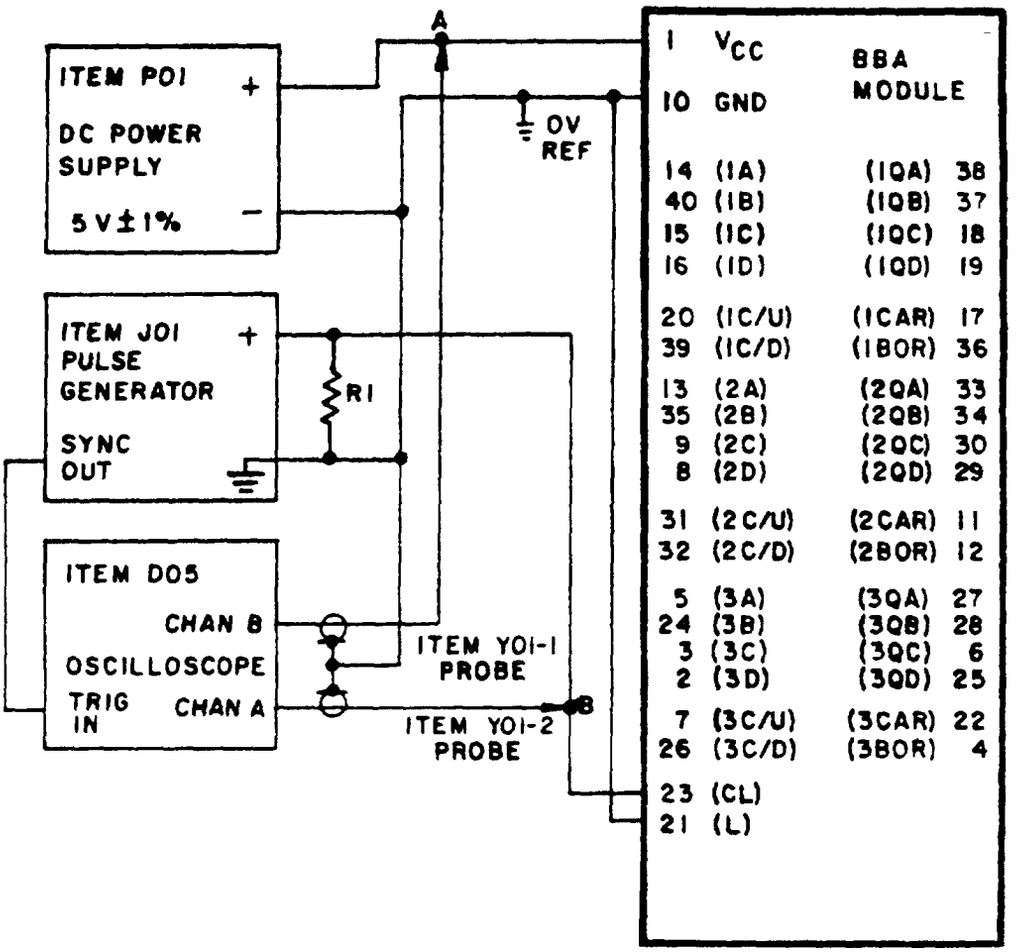
* FIGURE 26. Output crosstalk (duration) test circuit waveforms for table XVIII.

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TABLE XIX. Power supply transient amplitude test

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
DD	Power supply transient amplitude	0	400	mV	27,28

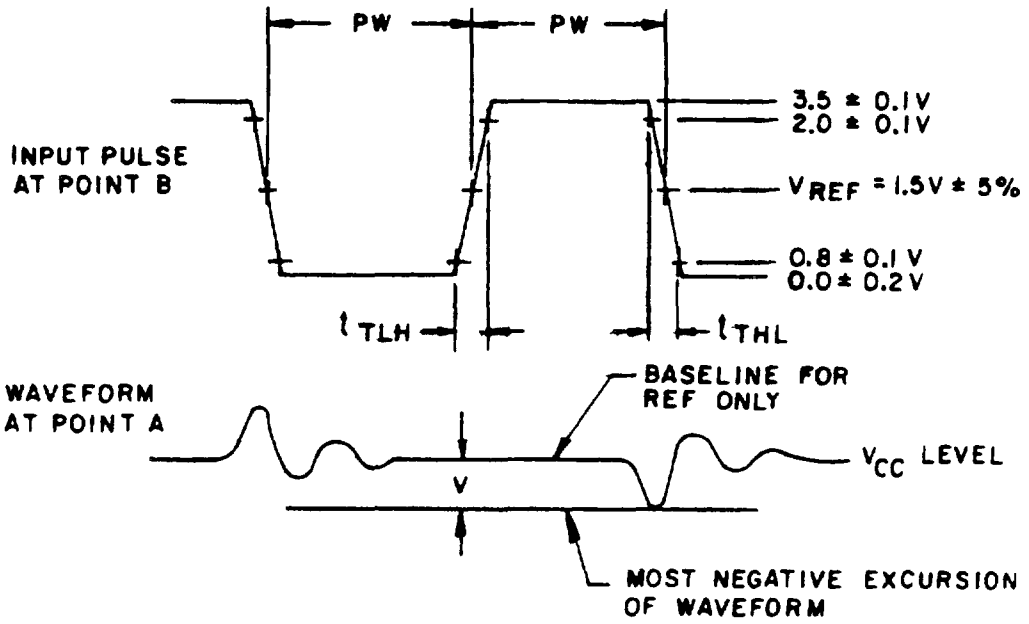
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NOTE: R1 = 51Ω ±5%, 1/2 W.

* FIGURE 27. Power supply transient amplitude test circuit for table XIX.

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NOTES:

1. Input pulse parameters:
   PW = 100 ±10 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
2. V = Measurement is made from the base line to the most negative excursion of the waveform.

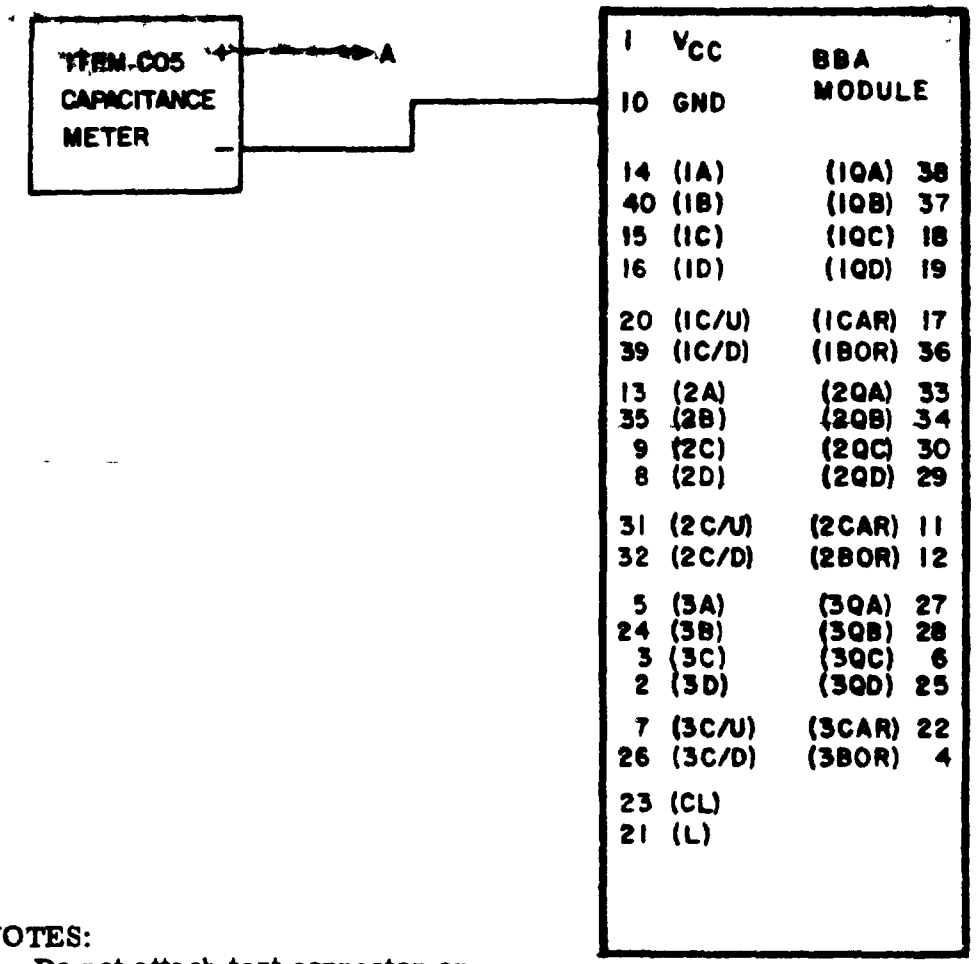
* FIGURE 28. Power supply transient amplitude test circuit waveforms for table XIX.

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TABLE XX. Input capacitance test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
EE1	1A Input capacitance	14	2	15	pF	29
EE2	1B "	40	"	"	"	"
EE3	1C "	15	"	"	"	"
EE4	1D "	16	"	"	"	"
EE5	1C/U "	20	"	"	"	"
EE6	1C/D "	39	"	"	"	"
EE7	2A "	13	"	"	"	"
EE8	2B "	35	"	"	"	"
EE9	2C "	9	"	"	"	"
EE10	2D "	8	"	"	"	"
EE11	2C/U "	31	"	"	"	"
EE12	2C/D "	32	"	"	"	"
EE13	3A "	5	"	"	"	"
EE14	3B "	24	"	"	"	"
EE15	3C "	3	"	"	"	"
EE16	3D "	2	"	"	"	"
EE17	3C/U "	7	"	"	"	"
EE18	3C/D "	26	"	"	"	"
EE19	CL "	23	4	30	"	"
EE20	L "	21	4	30	"	"

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 29. Input capacitance test circuit for table XX.

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